A Framework for Incentivizing Client Participation and Connectivity in Decentralized Federated Learning

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Abstract

In decentralized federated learning, edge clients collaboratively train a model over a network via peer-to-peer communication without relying on a central server. Each client may have its own requirements, such as achieving a minimum training loss, which it expects the learned model to satisfy. Due to data heterogeneity, these requirements might not be met, leading some clients to opt out of collaborative training. More critically, if certain highly connected clients drop out, it can lead to network fragmentation, thus disrupting the training process and causing the remaining clients to lose access to valuable information from other parts of the network. In this work, we introduce a novel framework designed to incentivize the continued participation of key clients, where the notion of "importance" can be defined using various criteria, such as data quality or connectivity metrics. We establish theoretical convergence guarantees for our proposed solution and validate our approach through empirical studies across various network topologies, demonstrating that our method outperforms standard decentralized learning under voluntary dropouts, achieving higher test-time accuracy while significantly improving client retention. Overall, our work yields accuracy improvements of up to 33 percentage points, with client retention rates reaching above 99% in the best cases.

CCS Concepts

• **Computing methodologies → Machine learning**.

Keywords

Decentralized Learning, Federated Learning, Client incentive mechanisms

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1 Introduction

Federated learning is a modern distributed machine learning paradigm [20, 30, 37] where edge clients collaboratively train a model using local datasets that cannot be transferred to the cloud for centralized training. In the standard implementation of federated learning a central server aggregates updates from edge devices. In systems where there is no central trustworthy aggregator the decentralized machine learning paradigm [23, 24, 31, 38, 39], nodes send and receive updates to and from neighboring nodes in a given network topology connecting them. Examples of such decentralized learning include banks, hospitals, or other entities that have local siloes of private data, but want to collaboratively learn models with other trusted entities. The network connecting the nodes then represents a trust graph used for communication with other nodes. Most previous works consider that all the clients participate in the network throughout the training process. In this paper, we study the problem of avoiding client dropouts, where nodes either voluntarily or involuntarily leave the collaborative training process.

Why would Clients Dropout from Decentralized Learning? In a fully decentralized learning framework in a connected network topology, the training datasets $\mathcal{D}_i$ across clients can be highly heterogeneous, a situation frequently encountered in practical applications [20]. Data heterogeneity results in the empirical risk objective functions $F_i(\mathbf{x})$ induced by the datasets being different across clients, and also different from the global objective function $\sum_{i=1}^{N} F_i(\mathbf{x})/N$. Each client maintains its own version of the model $\mathbf{x}_i$ that is trained for a few stochastic gradient descent (SGD) iterations on its local dataset and then averaged with neighboring clients. After several rounds of local training and neighbor aggregation in a connected topology, and under standard assumptions on the objective functions, all the nodes typically converge to a single collaboratively trained model $\mathbf{x}$. When the local data distribution is similar to the global distribution of the union of all the clients' datasets, the shared collaboratively trained model $\mathbf{x}$ generalizes better than a local model $\mathbf{x}_i$ trained by a client i in isolation using its limited local data $\mathcal{D}_i$. However, due to data heterogeneity, if the global distribution is significantly misaligned with a client's local data distribution, the client may prefer to leave the collaboration and train its own model in isolation.

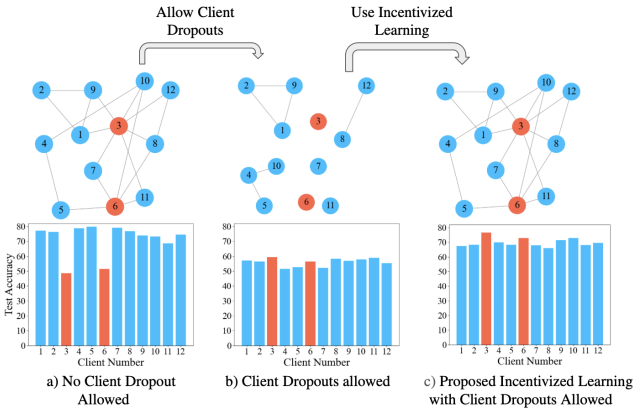


Figure 1: Test accuracies for a network of 12 clients under three algorithms: (a) standard decentralized learning without dropout yields low accuracy for clients 3 and 6; (b) allowing dissatisfied clients to drop out leads to network disconnection and accuracy loss; (c) the proposed incentivized framework retains key clients, preserving or improving accuracy for most clients.

Accuracy and Connectivity Disruption due to Client Dropout. When a client drops out, the other clients in the collaboration no longer get the generalization benefits of accessing a model trained on more diverse data. Recent works [8, 11, 12] have analyzed the effect of client dropout in a federated learning setting, where the remaining clients still remain connected with the central server. However, in a decentralized learning setting, since there is no central server, the departure of a node will also cause disconnection of its communication links to neighboring nodes. This can disrupt network connectivity, and cause other clients to get forcibly disconnected from the rest of the network. An extreme example is that of a star topology wherein losing the central node isolates all clients, reducing training to fully localized learning. In this paper, we propose a framework to incentivize connectivity-critical clients to continue participating in collaborative decentralized learning to provide generalization benefits to a majority of the clients and also maintain network connectivity.

Example Illustrating the Need to Retain Connectivity-Critical Clients. To illustrate the critical need to retain connectivity-critical clients in decentralized learning, and the effectiveness of our proposed incentivization framework, consider the example in Figure 1 of a network of 12 client nodes. Nodes 3 and 6 have disjoint label distributions compared to nodes 1, 2, 8, 9, 12 and nodes 4, 5, 10, 7, 11. Each client's test set consists of 60% data matching its training distribution and 40% data reflecting the distributions from other clusters. We compare the test accuracies of the clients with the following three different decentralized learning algorithms:

(a) **No Client Dropout Allowed.** The case in Figure 1 (a) corresponds to standard decentralized learning where all clients are required to participate in collaborative training, and implies that all clients must use the single global model $\mathbf{x}$ resulting from the collaboration. Since the model is optimized for the union of the datasets $\cup_i \mathcal{D}_i$, it performs well for most nodes but yields poor test accuracy for nodes 3 and 6, whose local data significantly differ from the global data distribution.

(b) **Client Dropouts Allowed.** For the results illustrated in Figure 1 (b), clients can opt-out if collaboration does not benefit them. The low accuracy experienced by nodes 3 and 6 in part (a) causes them to dropout and prefer to train their own models independently. However, the dropout of nodes 3 and 6 also disconnecting the rest of the network causing it to fragment into seven connected components, each of which train their own model. Since each client only learns from its connected peers, its model quickly loses exposure to diverse data; as a result, nodes 3 and 6 perform well on their specific label distributions, but the performance of all other nodes degrades once they no longer learn from the broader dataset.

(c) **Proposed Incentivized Learning with Client Dropouts Allowed.** To retain nodes 3 and 6, we propose an incentivized learning framework where we modify the objective function so that it penalizes poor accuracy of the shared model on each client's local data. The new objective also gives higher weight to the retention of connectivity-critical clients, losing whom will disrupt and fragment the rest of the clients in the network. This prevents network fragmentation and results in the shared model performing well on the local datasets of a majority of the clients. In Figure 1 (c) we observe that although other nodes experience a slight drop in performance due to the modification of the objective function, overall, all nodes perform better than in scenario (b), preserving network connectivity and improving global accuracy.

1.1 Our Contributions and Paper Organization

Client participation incentives have only been studied in a federated learning setting with a central aggregating server [8] rather than in a decentralized topology where voluntary dropout of some clients can force disconnection and fragmentation of the network. Thus, to our knowledge, this paper is the first consider the problem of incentivizing clients to participate in decentralized learning, with the purpose of maintaining the shared model's accuracy on clients' local data, while their local accuracy. The paper makes the following key contributions:

- In Section 2, we propose modified objective functions to incentivize key client participation by penalizing their dropouts: one penalizes dropouts of individual clients based on their importance, and other captures the collective importance of group of nodes by penalizing their simultaneous departure (e.g., nodes forming a vertex cut are collectively critical for connectivity).
- In Section 3, we propose `NodeDrop-IDSGD` that adapts client learning rates differently for each client to minimize the proposed dropout-aware objective.
- In Section 4, we establish theoretical convergence guarantees for our solution.
- In Section 5, we present weight-selection strategies that assign higher importance to structurally or statistically critical clients, using topology-based, data-based, and hybrid measures.
- In Section 6, we empirically evaluate our solution across various network topologies on image classification tasks. Our results show improved test accuracy (up to 33 percentage points) and greater

[0]Throughout this work we will use the terms client and node interchangeably

retention of nodes (up to 48 percentage points) in the collaboration network compared to standard decentralized learning, highlighting the critical importance of retaining key clients and preserving network connectivity for performance.

We omit a second, alternative proposed algorithm, as well as all proofs and experimental details, due to space limitations; these are provided in the appendix.

1.2 Related Work

In decentralized learning it is often more critical to retain nodes compared to federated learning since they also form the communication backbone. There are several empirical works that study the impact of losing a fraction of clients on performance in decentralized learning [17, 32, 34]. Beyond loss of data and possible network fragmentation, client dropouts can also affect the convergence rate by altering the network topology. There have been attempts to construct collaboration structure which helps attain faster convergence rates [3, 10, 27]. It has also been empirically demonstrated [33] that in scale-free networks, "central nodes" have a disproportionate impact on the final model performance and convergence rates. Collectively, these studies highlight that certain nodes play a more critical role than others in determining convergence-related metrics in fully decentralized learning. However they do not explicitly try to incentivize retention of such connectivity-critical clients.

Incentivizing client participation has been studied in federated learning, where a central server aggregates updates from all clients, but not in a decentralized topology. In the centralized federated setting, coalitional game theory has been used to model how self-interested agents form coalitions to maximize local utility ([11, 12]). Several works promote long-term client engagement by incorporating game-theoretic post-training reward mechanisms to incentivize participation ([16, 19, 21, 28, 42, 44]). Some approaches introduce training-time rewards to reduce dropouts [40] during training, while others focus on discouraging dishonest behavior, such as clients submitting large, manipulative updates [7, 13]. The most closely related work to ours changes the global objective function to make the model inherently appealing, encouraging more clients to remain in the collaboration [8]. This approach complements reward-based and personalization methods and can be integrated with them for further incentivization.

2 Problem Formulation

2.1 Notation

For any positive integer N, we define $[N] := \{1, \ldots, N\}$. The notation $\|\cdot\|$ or $\|\cdot\|_2$ represents the L_2-norm of a vector. When used for a matrix, $\|\cdot\|_F$ and $\|\cdot\|_2$ denote the Frobenius norm and the spectral norm, respectively. Lowercase boldface letters represent vectors, while uppercase boldface letters represent matrices. The symbol $\mathbb{I}(\cdot)$ denotes the indicator function, and $\mathbf{1}$ represents an all-ones vector of appropriate dimension.

2.2 Decentralized SGD Preliminaries

We consider a scenario where N clients (or nodes) collaborate via a connected network topology to train a shared model parameterized by $\mathbf{x} \in \mathbb{R}^d$. Each client i holds a local dataset $\mathcal{D}_i$, consisting of samples drawn from its respective underlying true data distribution Z_i. The true loss function for client i, given by $f_i(\mathbf{x}) = \mathbb{E}_{\xi \sim Z_i}[l(\mathbf{x}, \xi)]$ and is approximated by the empirical loss $F_i(\mathbf{x}) = \frac{1}{|\mathcal{D}_i|} \sum_{\xi \in \mathcal{D}_i} l(\mathbf{x}, \xi)$, as per the standard empirical risk minimization approach. The N nodes seek to collaboratively minimize the average of the local empirical loss functions, $\sum_{i=1}^{N} F_i(\mathbf{x})/N$, which we sometimes refer to as the global objective function.

Algorithm 1 DECENTRALIZED SGD WITH LOCAL UPDATES

Require: For each node $i \in [N]$, initialize $\mathbf{x}_i^{(0,0)} \in \mathbb{R}^d$, stepsizes $\{\eta_t\}_{t=0}^{T-1}$, no. of iterations T, mixing matrix $\mathbf{W}$

for $t \in 0 \ldots T$ **do**
  For each worker $i, i \in [N]$**:**
    For $r = 0, 1 \cdots (\tau - 1)$ **local iterations**
      Sample mini-batch of data $\xi_i^{(t,r)}$,
      Compute local gradient $\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})$
      $\mathbf{x}_i^{(t,r+1)} := \mathbf{x}_i^{(t,r)} - \eta_t \nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})$; local updates
    $\mathbf{x}_i^{(t+1,0)} := \sum_{j \in \mathcal{N}_i} w_{ij} \mathbf{x}_j^{(t,\tau)}$; gossip averaging
end for

The N nodes (or clients) communicate through an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{1, \ldots, N\}$ represents the set of nodes, and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ denotes the set of undirected edges connecting them. Each edge $(i, j) \in \mathcal{E}$ signifies a bidirectional communication link between nodes i and j. The neighborhood of a node i is given by $\mathcal{N}_i = \{j \mid (i, j) \in \mathcal{E}\}$, representing all nodes directly connected to i.

The standard decentralized SGD algorithm with local updates is given in Algorithm 1. We denote the local model at the starting of round t at node i by $\mathbf{x}_i^{(t,0)}$. Within every round t each client executes τ steps of mini-batch stochastic gradient descent and the intermediate model at node i after $0 \le r \le \tau$ are denoted by $\mathbf{x}_i^{(t,r)}$. We define the vector of these models

$$\mathbf{X}^{(t,0)} := \left[\mathbf{x}_1^{(t,0)}, \ldots, \mathbf{x}_N^{(t,0)}\right] \in \mathbb{R}^{d \times N}.$$

After local gradient updates at each nodes the local models are updated using weighted average of the neighboring models as $\mathbf{X}^{(t+1,0)} = \mathbf{X}^{(t,\tau)}\mathbf{W}$, where $\mathbf{W} = [w_{ij}]_{i,j \in [N]} \in [0, 1]^{N \times N}$ is referred to the mixing matrix. This averaging step is equivalent to each node performing a weighted average of the neighboring models $\mathbf{x}_i^{(t+1,0)} = \sum_{j \in \mathcal{N}_i} w_{ij} \mathbf{x}_j^{(t,\tau)}$. We assume that the mixing matrix is symmetric ($\mathbf{W} = \mathbf{W}^\top$) and doubly stochastic ($\mathbf{W}\mathbf{1} = \mathbf{1}, \mathbf{1}^\top \mathbf{W} = \mathbf{1}$), satisfies $w_{ij} = 0$ if $(i, j) \notin \mathcal{E}$ and $w_{i,i} \neq 0 \ \forall i \in [N]$. Decentralized SGD relies on the key idea that when the learning rates are sufficiently small, the effect of gossip averaging dominates over local updates, leading to consensus among the models across nodes. The convergence of this algorithm has been analyzed in [23].

2.3 Client Dropout Model

In standard decentralized learning, it is typically assumed that all clients take part in the training process throughout its duration. However, clients whose local datasets differ significantly from the global data distribution often suffer from degraded performance and may want to opt-out of collaborative training if allowed (see Figure 1).

Here we will describe a client dropout model that can be used to analyze this behavior.

Each client i seeks a model $\mathbf{x}$ that minimizes its true loss function $f_i(\mathbf{x}) = \mathbb{E}_{\xi\sim Z_i}[l(\mathbf{x},\xi)]$. If $\mathbf{x}_i^{(T,0)}$ denote the model obtained after T rounds, the client has a self-defined threshold ρ_i on its loss $f_i(\mathbf{x}_i^{(T,0)})$ to judge and decide whether to stay in the system. If $f_i(\mathbf{x}_i^{(T,0)}) > \rho_i$ the client decides to dropout and instead choose to train a model on its own in isolation. A good proxy for ρ_i can be obtained by the client i by a few rounds of local training of the model on a held-out dataset. Since the true data distribution Z_i is unknown to the client i, it replaces the true loss $f_i(\mathbf{x}_i^{(T,0)})$ with the empirical loss $F_i(\mathbf{x}_i^{(T,0)})$. Thus if $\mathcal{Y}_i$ as the event that client i drops out,

$$\mathbb{I}(\mathcal{Y}_i) = \mathbb{I}(F_i(\mathbf{x}_i^{(T,0)}) > \rho_i) \tag{1}$$

$$\approx \sigma_i(\mathbf{x}_i^{(T,0)}) \triangleq \frac{1}{1+\exp(-(F_i(\mathbf{x}_i^{(T,0)}) - \rho_i))} \tag{2}$$

which is the sigmoid function $F_i(\mathbf{x}_i^{(T,0)}) - \rho_i$. Since the indicator function is non-differentiable, in (2), we approximate it using the sigmoid function, a standard approximation employed in optimization literature.

Remark 1 (Probabilistic interpretation Using the Bradley-Terry Model). *The sigmoid approximation of the indicator admits a probabilistic interpretation, where $\sigma_i(\mathbf{x}_i^{(T,0)})$ can be viewed as the probability that client i drops out, and it aligns with the well-known Bradley-Terry model ([5]) used for pairwise comparisons. As per the Bradley-Terry model, for a score function $r(\mathbf{x})$ indicative of the utility of object $\mathbf{x}$, the probability of preferring an object $\mathbf{x}_a$ over $\mathbf{x}_b$ is expressed as:*

$$\Pr(\mathbf{x}_a \succ \mathbf{x}_b) = \frac{\exp(r(\mathbf{x}_a))}{\exp(r(\mathbf{x}_a)) + \exp(r(\mathbf{x}_b))}.$$

Defining the score for a model as the negative of the loss, $r(\mathbf{x}_i^{(T,0)}) = -F_i(\mathbf{x}_i^{(T,0)})$, and letting ρ_i represent client i's estimate of the achievable loss from local training, giving it a score $r(\mathbf{x}_i^{local}) = -\rho_i$, the probability that client i opts out or prefers the local model over the global model can be expressed as:

$$\Pr(\textit{Client i drops out}) = \Pr(\mathbf{x}_i^{local} \succ \mathbf{x}_i^{(T,0)})$$
$$= \frac{\exp(-\rho_i)}{\exp(-\rho_i) + \exp(-F_i(\mathbf{x}_i^{(T,0)}))} = \sigma_i(\mathbf{x}_i^{(T,0)}).$$

This probabilistic interpretation providing a strong theoretical grounding to the proposed dropout model to complement its practical applicability.

2.4 Proposed Modified Objective Functions

Standard decentralized SGD over a connected network topology optimizes the average of the local objective functions, $\sum_{i=1}^N F_i(\mathbf{x})/N$, to obtain a shared model $\mathbf{x}$ that will be later used by the clients in the network. However, due to data heterogeneity, clients whose local data $\mathcal{D}_i$ differs greatly from the union of the datasets $\cup_i \mathcal{D}_i$ will suffer from poor accuracy, as illustrated in Figure 1 part (a). If these dissatisfied clients are allowed to dropout, then it causes accuracy loss for many other clients due to disruption of the connected network topology. We propose the following two ways to modify the objective function: node-wise penalty and group-wise penalty, to encourage connectivity-critical clients to continue participating in the collaboration. Node-wise penalty assigns weights to each node according to that node's local properties, such as its degree, that impact the network's connectivity. Group-wise penalty considers the effect of the departure of groups of nodes on the connectivity.

Modified Objective with Node-wise Penalty. To minimize client dropouts and maximize the number of clients whose performance requirements are met by the resulting model, [8] propose minimizing the objective function given by

$$\tilde{F}(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N \mathbb{I}(F_i(\mathbf{x}) > \rho_i) \approx \frac{1}{N}\sum_{i=1}^N \sigma_i(\mathbf{x}). \tag{3}$$

This objective function assigns equal importance to each node. However, as discussed earlier, in fully decentralized learning, the importance of nodes is dependent on topology. Thus, it becomes increasingly important to retain key nodes in order to sustain efficient collaboration. To account for this disparity, we propose using a generalized weighting scheme that can incorporate importance weights a_i. We define our modified objective function $\tilde{F}(\mathbf{x})$ as follows

$$\tilde{F}(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N \tilde{F}_i(\mathbf{x}) := \frac{1}{N}\sum_{i=1}^N a_i\sigma_i(\mathbf{x}) \tag{4}$$

where a_i represents the relative importance of node i. It can be thought of as the penalty for node i leaving the system. For example, one practical choice of a_i is the degree of the node i. From the Bradley-Terry probabilistic interpretation in Remark 1, one can think of $\tilde{F}$ as the expected penalty incurred from dropouts under a model $\mathbf{x}$.

More generally, we can interpolate between minimizing average empirical loss and penalizing key-client dropouts by introducing a mixing parameter $\gamma \in [0,1]$

$$G(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N G_i(\mathbf{x}) = (1-\gamma)F(\mathbf{x}) + \gamma\tilde{F}(\mathbf{x}) \tag{5}$$

$$\triangleq (1-\gamma)\left(\frac{1}{N}\sum_{i=1}^N F_i(\mathbf{x})\right) + \gamma\left(\frac{1}{N}\sum_{i=1}^N a_i\sigma_i(\mathbf{x})\right). \tag{6}$$

When $\gamma = 0$ this reduces to the standard average empirical loss minimization objective. Setting $\gamma = 1$ and $a_i = 1\ \forall i$ recovers the objective function proposed in [8] for incentivizing client participation in the centralized federated learning setting. Notably, [8] shows that incentivizing more clients to participate in training helps the model generalize effectively to unseen inference-time clients who do not participate in collaborative training.

Modified Objective with Group-wise Penalty. In many real-world scenarios, the importance of a node is not solely determined in isolation but rather depends on its relationship with other nodes. Often, groups of nodes collectively contribute to the system's overall stability and function. For example, consider the scenario where nodes a, b, c together form a minimal vertex cut: a critical subset of nodes whose joint removal disconnects the network. If all three nodes were to leave the system, they would fragment the collaboration structure into two disjoint sets, completely severing communication between them. However, if only a subset of these nodes were to leave, the graph would remain connected. In this scenario, we want

to associate a penalty with the joint dropout of an entire group of critical nodes. Let $\mathcal{Y}_i$ represent the event that node i leaves the system. To capture the impact of a complete group departure, we introduce a penalty linked to the event:

$$\mathbb{I}(\mathcal{Y}_a \cap \mathcal{Y}_b \cap \mathcal{Y}_c) = \mathbb{I}(\mathcal{Y}_a)\mathbb{I}(\mathcal{Y}_b)\mathbb{I}(\mathcal{Y}_c) \approx \sigma_a(\mathbf{x}_a^{(T,0)})\sigma_b(\mathbf{x}_b^{(T,0)})\sigma_c(\mathbf{x}_c^{(T,0)})$$

that all three nodes a, b, c simultaneously exit the system, thereby disrupting network connectivity. If only a subset of these nodes leaves the overall network remains connected and thus no penalty is applied. In general, let there be K groups of 'collectively important' nodes where we denote the kth group of nodes by C_k. We define the group-wise penalty function

$$\tilde{F}(\mathbf{x}) = \frac{1}{N}\sum_{k=1}^{K}\lambda_k \prod_{j\in C_k}\sigma_j(\mathbf{x}) \tag{7}$$

where the λ_k is the weight assigned to group C_k. From the Bradley Terry interpretation in Remark 1, the product $\prod_{j\in C_k}\sigma_j(\mathbf{x})$ can be thought of as the probability of all nodes within the group C_k leaving the system. This is because given the model $\mathbf{x}$, each node independently decides whether to stay or drop out of collaboration, and thus, $\mathcal{Y}_i$ are independent events.

We again define our generalized objective function by interpolating between the empirical risk minimization and group-wise penalty function:

$$G(\mathbf{x}) = (1-\gamma)F(\mathbf{x}) + \gamma\tilde{F}(\mathbf{x}) = \frac{1}{N}\left\{(1-\gamma)\left(\sum_{i=1}^{N}F_i(\mathbf{x})\right) + \gamma\left(\sum_{k=1}^{K}\lambda_k\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)\right\}. \tag{8}$$

Observe that the modified objective function in (8) with the group-wise penalty subsumes the objective in (6) with node-wise penalty by including groups C_k where $|C_k| = 1$ to represent individual nodes.

3 Proposed Incentivized Learning Algorithm

We propose an incentivized decentralized learning algorithm for the modified objective function in (6). For the modified objective function in (8), the alternative proposed algorithm is in Appendix A due to space limit.

3.1 `NodeDrop-IDSGD` for the Node-wise Penalty Objective

We introduce `NodeDrop-IDSGD`, an algorithm designed to minimize the objective function with node-wise penalty, as defined in Equation (6). Notably, `NodeDrop-IDSGD` does not alter the local model update procedure from Algorithm 1, ensuring that the core optimization framework remains unchanged. At the start of each round $t \in T$, the only additional step in `NodeDrop-IDSGD` is the estimation of an adaptive learning rate for each user (lines 3-6). Once this learning rate is computed, users proceed with the standard τ local stochastic gradient descent (SGD) updates, followed by a gossip averaging step to mix models with their neighbors. Since `NodeDrop-IDSGD` modifies only the learning rate at each user while keeping the rest of the procedure unchanged, it remains a simple yet effective extension of decentralized SGD.

Algorithm 2 `NodeDrop-IDSGD`

Require: For each node $i \in [N]$, initialize $\mathbf{x}_i^{(0,0)} \in \mathbb{R}^d$, base LR $\{\eta\}_{t=0}^{T-1}$, no. of iterations T, mixing matrix $\mathbf{W}$

1: **for** $t \in 0 \dots T$ **do**
2: **For each worker** $i, i \in [N]$**:**
3: **`Learning Rate Estimation Phase:`**
4: Compute $q_i(\mathbf{x}_i^{(t,0)}) = \sigma(\mathbf{x}_i^{(t,0)})(1-\sigma(\mathbf{x}_i^{(t,0)}))$
5: Calculate Pre-factor $b_i^{(t)} = (1-\gamma) + \gamma a_i q_i(\mathbf{x}_i^{(t,0)})$
6: Compute scaling factors $s_i^{(t)}$ using gossip (τ_η steps). `// scalar gossip averaging`
7: Set effective learning rate $\hat{\eta}_i^{(t)} = \frac{\eta b_i^{(t)}}{\tau(s_i^{(t)}+\epsilon)}$.
8: **`Model Update Phase:`**
9: **For** $r = 0, 1 \cdots (\tau-1)$ **local iterations**
10: Sample $\xi_i^{(t,r)}$,
11: Compute $\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})$
12: $\mathbf{x}_i^{(t,r+1)} := \mathbf{x}_i^{(t,r)} - \hat{\eta}_i^{(t)}\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})$
13: $\mathbf{x}_i^{(t+1,0)} := \sum_{j\in N_i} w_{ij}\mathbf{x}_j^{(t,\tau)}$ `// model gossip averaging`
14: **end for**

The gradient of the modified objective function G is a weighted average of the gradients of the empirical loss functions F_i given by

$$\nabla G(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^{N}\left((1-\gamma) + \gamma a_i \underbrace{\sigma(\mathbf{x})(1-\sigma(\mathbf{x}))}_{q_i(\mathbf{x})}\right)\nabla F_i(\mathbf{x}).$$

We denote the pre-factor that is multiplied to the gradient $\nabla F_i(\mathbf{x})$ at node i at time t by

$$b_i^{(t)} = (1-\gamma) + \gamma a_i q_i(\mathbf{x}_i^{(t,0)})$$

where, according to our client dropout model, $q_i(\mathbf{x})$ takes a larger value if the client is dissatisfied with model $\mathbf{x}$ and its dropout probability is close to 0.5.

If we were to apply simple gradient descent to minimize G_i, the update rule would be given by $\mathbf{x}_i^{(t+1)} \leftarrow \mathbf{x}_i^{(t)} - \eta\nabla G_i(\mathbf{x}_i^{(t)}) = \mathbf{x}_i^{(t)} - \eta b_i^{(t)}\nabla F_i(\mathbf{x}_i^{(t)})$. Here, η can be interpreted as the learning rate with respect to G_i, while $\eta b_i^{(t)}$ serves as the effective learning rate with respect to F. Multiplication by the pre-factor can be used to account for the relation between the gradients of G and F.

Learning Rate Estimation Phase: The process starts with a simple inference step (line 4) wherein each client computes $q_i(\mathbf{x}_i^{(t,0)}) = \sigma(\mathbf{x}_i^{(t,0)})(1-\sigma(\mathbf{x}_i^{(t,0)}))$. In Algorithm 2, $\hat{\eta}_i^{(t)}$ can be interpreted as the learning rate with respect to F, while $\frac{\eta}{\tau(s_i^{(t)}+\epsilon)}$ corresponds to the learning rate with respect to G. The normalization by τ accounts for the number of local updates, is simply a matter of convention and does not affect our results. The term $s_i^{(t)}$ denotes the scaling factor at client i used to approximate the global sum of pre-factors $s^{(t)} := \sum_{i=1}^{N} b_i^{(t)}$ for normalization. This results in the final learning rate at each client: $\hat{\eta}_i^{(t)} = \frac{\eta b_i^{(t)}}{\tau(s_i^{(t)}+\epsilon)}$. The smoothness parameter of the global objective function's (G) landscape ($\tilde{L}_s$) depends on the

point $\mathbf{x}$ and takes the form $A + B\sum_{i=1}^{N}((1-\gamma) + \gamma a_i q_i(\mathbf{x}))$, as shown in Lemma 2 of the Appendix C).

Following standard practice in stochastic optimization, we adapt the learning rate with respect to G to be $\frac{1}{\tilde{L}_s} = \frac{1}{A+B\sum_{i=1}^{N} b_i^{(t)}}$ This dynamic normalization is incorporated into our algorithm by setting the learning rate for G to $\frac{\eta}{\epsilon+\sum_{i=1}^{N} b_i^{(t)}}$, where η and ϵ are tunable hyperparameters, and the denominator acts as a dynamic normalization factor. Lastly note that we compute this learning rate estimation procedure only once for every τ local SGD steps.

Estimating Scaling Factors: The scaling factors $s_i^{(t)}$ are estimated using gossip averaging with the same mixing matrix $\mathbf{W}$. Each user i only has access to its local pre-factor $b_i^{(t)}$ and direct communication between all nodes is impractical in a decentralized setup. To address this we use gossip averaging, which enables nodes to iteratively exchange information with their neighbors to collaboratively approximate $s^{(t)}$. To estimate $s^{(t)}$, each user participates in τ_η steps of gossip averaging, initializing the process with $s_i^{(t,0)} = Nb_i^{(t)}$. This ensures that each node starts with a scaled version of its local term before iteratively refining its estimate. During gossip averaging, the scaling factors are iteratively updated using the rule: $s_i^{(t,r_\eta+1)} = \sum_{j\in\mathcal{N}_i} w_{ij} s_j^{(t,r_\eta)}$ which can be expressed in matrix form as: $\mathbf{s}^{(t,r_\eta+1)} = \mathbf{s}^{(t,r_\eta)}\mathbf{W}$, with $\mathbf{s}^{(t,r_\eta)} = [s_1^{(t,r_\eta)} \cdots s_N^{(t,r_\eta)}]$ for $0 \le r_\eta \le \tau_\eta - 1$. After τ_η steps of gossip averaging node i estimates $s^{(t)}$ as

$$s_i^{(t)} = \max\left\{s_i^{(t,\tau_\eta)}, s_i^{(t,0)}/N\right\} = \max\left\{s_i^{(t,\tau_\eta)}, b_i^{(t)}\right\}.$$

This step ensures that the estimate remains robust and non-degenerate, preventing cases where averaging might lead to values that are too small. By enforcing a lower bound of $b_i^{(t)}$ we guarantee that the estimate of the sum the pre-factors of all nodes $\in [N]$ at node i is larger than the pre-factor at node i ($b_i^{(t)}$). Thus finally client i approximates the effective learning rate $\left(\frac{\eta b_i^{(t)}}{\tau(s^{(t)}+\epsilon)}\right)$ using $\hat{\eta}_i^{(t)} = \frac{\eta b_i^{(t)}}{\tau(s_i^{(t)}+\epsilon)}$.

REMARK 2 (MINIMAL COMMUNICATION OVERHEAD FOR SCALING FACTOR ESTIMATION). *The τ_η gossip averaging steps used to estimate the scaling factor. Since the scaling factor is a one-dimensional quantity it incurs a negligible communication cost compared to the model averaging step, which involves exchanges of high-dimensional deep neural network parameters (line 12 of Algorithm 2).*

Model Update Phase: Given the modified learning rate the model update proceeds in the same way as decentralized SGD (Algorithm 1). For the rth local step in round t each client i independently samples a mini-batch $\xi_i^{(t,r)}$, computes the gradient w.r.t. $\xi_i^{(t,r)}$ and updates the model parameters.

4 Theoretical Guarantees

We will discuss the convergence guarantees for our proposed solution. We start by listing some assumptions commonly used in federated, decentralized learning and optimization literature. [22–24, 35, 36]. Note that we will not make any assumptions about our modified objective functions; rather, all our assumptions are limited to the local empirical loss functions F_i.

ASSUMPTION 1. *(Unbiased Stochastic Gradient with Bounded Variance) : For any mini batch ξ_i sampled uniformly from the dataset we have the expected value of the stochastic gradient $\mathbb{E}[\nabla F_i(\mathbf{x}, \xi_i)] = \nabla F_i(\mathbf{x})$ and its variance bounded as $\mathbb{E}[\|\nabla F_k(\mathbf{x}_k, \xi_k) - \nabla F_k(\mathbf{x}_k)\|^2] \le \sigma^2$.*

ASSUMPTION 2. *(Continuity & Smoothness of local objective functions) : The local objective functions $F_1(\mathbf{x}), \ldots, F_N(\mathbf{x})$, are L_c-continuous and L_s-smooth for any $\mathbf{x}$.*

ASSUMPTION 3. *(Bounded Dissimilarity of $F(\mathbf{x})$) : There exist constants κ and β such that*

$$\frac{1}{N}\sum_{i=1}^{N}\|\nabla F_i(\mathbf{x})\|_2^2 \le \kappa^2 + \beta^2\|\frac{1}{N}\sum_{i=1}^{N}\nabla F(\mathbf{x})\|_2^2.$$

ASSUMPTION 4. *(Mixing matrix and Consensus Rate) We assume that the mixing matrix is symmetric ($\mathbf{W} = \mathbf{W}^\top$) and doubly stochastic ($\mathbf{W1} = \mathbf{1}, \mathbf{1}^\top\mathbf{W} = \mathbf{1}$), satisfies $w_{ij} = 0$ if $(i,j) \notin \mathcal{E}$ and $w_{i,i} \neq 0$ $\forall i \in [N]$. In addition, we assume that there exists a constant $p \in (0,1]$ such that for any $\mathbf{X} \in \mathbb{R}^{d\times N}$ we have $\|\mathbf{XW} - \bar{\mathbf{X}}\|_F^2 \le (1-p)\|\mathbf{X} - \bar{\mathbf{X}}\|_F^2$. Here $\bar{\mathbf{X}} = \mathbf{X}\frac{\mathbf{11}^T}{N}$.*

This property always holds for p equal to the second largest eigenvalue of the matrix W given by $\left\|\mathbf{W} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top\right\|_2$. A larger p means better connectivity and more efficient information mixing, p for a fully connected/federated topology is 1. The convergence is typically analyzed in terms of an average virtual model at the start of round t, denoted by

$$\bar{\mathbf{x}}^{(t,0)} := \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i^{(t,0)}.$$

THEOREM 1 (CONVERGENCE OF NODEDROP-IDSGD). *For the objective function G with nodewise penalty defined in Equation (6), under assumptions 1-4 there exists a base learning rate η under which for Algorithm 2 we have*

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G(\bar{\mathbf{x}}^{(t,0)})\right\|^2\right] \le \mathcal{O}\left(\sqrt{\frac{G_{\text{diff}}(1+\epsilon_\eta)}{N\tau T}}\right) + \mathcal{O}\left(\frac{(G_{\text{diff}})^{2/3}(1+\epsilon_\eta^{1/3})}{p^{2/3}T^{2/3}}\right) + \mathcal{O}\left(\frac{(G_{\text{diff}})^{2/3}(1+\epsilon_\eta^{1/3})}{p^{2/3}\tau^{1/3}T^{2/3}}\right) + \mathcal{O}\left(\frac{\Xi_0^2}{pT}\right) + \mathcal{O}\left(\frac{G_{\text{diff}}}{T}\right) + \mathcal{O}\left(\epsilon_\eta\right).$$

Here $G_{\text{diff}} = G(\bar{\mathbf{x}}^{(0,0)}) - G_{\text{inf}}$, is the initial distance from the infimum and $\mathcal{O}$ subsumes all constants. The parameter τ denotes the number of local steps performed in each iteration. The term $\epsilon_\eta = \frac{N^2(1-p)^{\tau_\eta}}{4}$ represents the learning rate estimation error, where τ_η is the number of gossip averaging steps used for estimating the scaling factors. The initial consensus error defined as $\Xi_0^2 = \frac{1}{N}\sum_{i=1}^{N}\|\bar{\mathbf{x}}^{(0,0)} - \mathbf{x}_i^{(0,0)}\|^2$.

The proof is deferred to Appendix C. The only term that does not vanish as $T \to \infty$ is the learning rate estimation error $\epsilon_\eta = \frac{N^2(1-p)^{\tau_\eta}}{4}$. This term, however, can be made arbitrarily small by increasing τ_η, independent of the other terms with negligible communication overhead (see Remark 2). We observe a linear speedup in both the number of clients N and the number of local steps τ for the dominant term ($\mathcal{O}(T^{-1/2})$), which decays the slowest with T.

5 Measures of Node Importance

In `NodeDrop-IDSGD`, the importance weights a_i control how strongly the algorithm penalizes the dropout of client i. Clients with larger a_i incur a higher penalty when they leave, translating into larger effective learning rates and a stronger incentive to remain in the collaboration. The choice of a_i is therefore the mechanism by which we encode which clients are most critical for accuracy and connectivity. In real-world scenarios, we often have some prior knowledge about the importance of retaining node i in the system. This importance may reflect the amount or quality of data the node contributes, or the cost and complexity of restoring network connectivity if the node is lost. Such information can serve as a useful proxy when selecting the values of a_i to incorporate the cost of losing client i into the objective function.

We consider a broad family of weighting rules $a = (a_1, \ldots, a_N)$, all implemented through a common interface that takes as input the communication graph G. The strategies we evaluate fall into three classes: topology-based, data-based, and hybrid.

5.1 Importance Derived from Topology

Topology-based strategies assign importance using only the structure of the communication graph. We consider the following choices:

Degree. [15] Assigns larger a_i to high-degree nodes ($a_i = \deg(i)$) that offer more alternative paths and faster gossip mixing.

Degree (normalized). [15] Uses a normalized degree, obtained by dividing each node's degree by the average degree of the graph, to keep a_i in a reasonable range preserving the intuition of Degree while avoiding excessively large weights on high-degree nodes.

Betweenness [14]. Sets a_i proportional to how often node i lies on shortest paths between other nodes, prioritizing clients that mediate many communication routes and protecting connectivity bottlenecks.

Eigenvector [4]. Assigns larger a_i to clients that are connected to other highly weighted clients, capturing influence in dense core regions rather than just local degree.

Closeness [15]. Sets a_i according to how close node i is to all other nodes, favoring clients that can quickly reach the rest of the network, accelerating information propagation.

PageRank [6]. Defines a_i based on a random-walk importance score, assigning larger weights to nodes visited frequently by that walk.

Vertex connectivity. Assigns larger a_i to nodes whose removal causes the largest drop in global vertex connectivity. Sets $a_i = 1 + \big(\kappa(G) - \kappa(G \setminus \{i\})\big)$, where κ denotes the vertex connectivity of the graph and $G \setminus \{i\}$ is the graph obtained by removing node i.

All centrality-based importance strategies mentioned before are computed using the corresponding implementations in NetworkX [18].

5.2 Importance Derived from Data Distribution

Data-based strategies ignore the graph structure and derive a_i from the empirical label distribution p_i at client i:

KL Divergence. Sets $a_i = 1 + \mathrm{KL}(p_i \,\|\, p_{\text{global}})$, where p_{global} is the average label distribution across clients and KL is the Kullback–Leibler divergence [26] (a non-negative measure of how different two discrete distributions are). This assigns larger weights to clients whose data deviate most from the global distribution, thereby prioritizing sources of rare or non-representative data.

Inverse Entropy. Sets $a_i = 1 + (H_{\max} - H(p_i))/H_{\max}$, where $H(p_i)$ is the entropy of the local label distribution and $H_{\max}$ its maximum value. Clients with low entropy (highly skewed data) receive larger weights, reflecting their role in representing rare labels. Clients with uniform data are treated as less critical even if they are numerous.

Dropout Risk (DR). Combines divergence and inverse entropy into a single score $a_i = 1 + \mathrm{KL}(p_i \,\|\, p_{\text{global}})\big(1 - H(p_i)/H_{\max}\big)$. Clients are emphasized when they are both far from the global distribution and highly specialized. This targets exactly the nodes that are expected to leave and that are most costly to lose.

The constant 1 added for these strategies ensures that all clients receive a positive baseline weight while the additional term increases the weight of more specialized or divergent clients.

REMARK 3. *In these data-imbalance scenarios, our modified objective functions can also be applied in a traditional (centralized) federated learning setting. When a central server manages aggregation, the same importance weights a_i can be used without incurring the estimation errors that arise in a fully decentralized coordination.*

5.3 Hybrid Importance Measures

Hybrid strategies combine structural and statistical signals, assigning higher importance to clients that are simultaneously critical for connectivity and for representing heterogeneous data:

Degree with KL. Sets $a_i = \deg(i)\big(1 + \mathrm{KL}(p_i \,\|\, p_{\text{global}})\big)$, emphasizing clients that are both highly connected and whose label distribution differs substantially from the global one. of the graph.

Betweenness with KL. Sets a_i to betweenness scores scaled by $1 + \mathrm{KL}(p_i \,\|\, p_{\text{global}})$, so that clients lying on many shortest paths and also exhibiting heterogeneous label distributions receive the largest a_i prioritizing structurally critical nodes whose dropout would both fragment the graph and remove access to rare data labels.

Betweenness with DR. Sets a_i to betweenness-based scores scaled by the Dropout Risk score, thereby prioritizing nodes that are both central to many shortest paths and highly specialized and far from the global distribution.

Eigenvector with KL. Sets a_i to eigenvector scores scaled by $1 + \mathrm{KL}(p_i \,\|\, p_{\text{global}})$, emphasizing clients that are embedded in influential core regions of the graph and whose data are atypical.

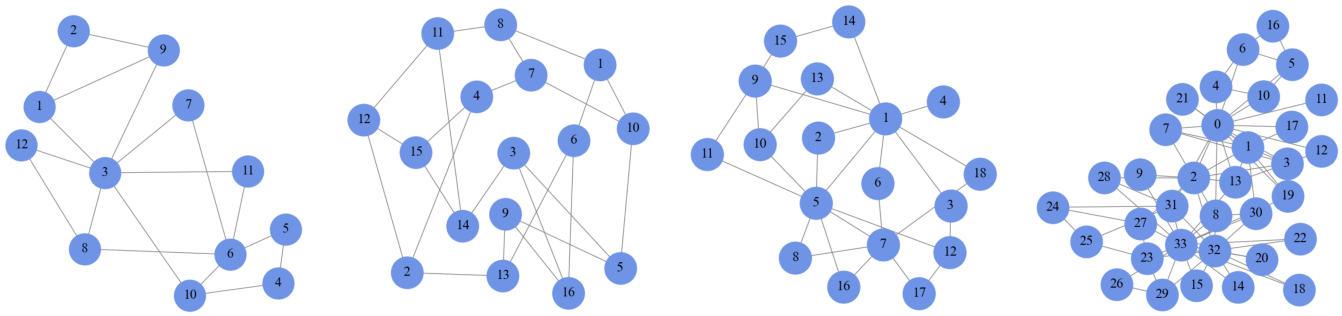
PageRank with KL. Sets a_i to a random-walk-based importance score scaled by $1 + \mathrm{KL}(p_i \,\|\, p_{\text{global}})$, emphasizing nodes that are frequently visited by random walks and whose label distributions differ substantially from the global one.

6 Experiments

All the experiments are done on the following graph topologies, illustrated in Figure 2.

(1) **Scale Free Like 12 (SFL12).** A network of 12 clients with node degrees sampled to roughly follow a power-law distribution, mimicking the structure of a scale-free network [2]. In such networks, a few nodes act as hubs with high connectivity, while most have low degrees, reflecting heterogeneity often seen in real-world systems like social or communication networks.

(2) **16 Node 3 Connections (3Con16).** A uniform network of 16 clients where each node has exactly 3 connections (degree 3). The network is harder to fully disconnect due to its regular and balanced connectivity.



(a) Scale Free Like 12 **(b) 16 Node 3 Connections** **(c) Scale Free Like 18** **(d) Karate Club Network**

Figure 2: Graph topologies considered in our experiments.

(3) **Scale Free Like 18 (SFL18)** - An 18-client extension of the SFL 12 network, preserving the general shape of a power-law-inspired degree distribution.
(4) **Karate Club Network (KCN)** - A well-known 34-node social network ([43]) with a few high degree nodes, It features a skewed degree distribution with an average degree of 4.59, a maximum of 17, and a minimum of 1.

The mixing weights are chosen according to Metropolis-Hastings mixing matrix construction where $w_{i,j} = \frac{1}{1+\max\{\deg(i),\deg(j)\}}$ for $i \neq j$, $(i,j) \in \mathcal{E}$ and $w_{i,i} = 1 - \sum_{j \in \mathcal{N}_i} w_{i,j}$. Note that this gives us a symmetric and doubly stochastic matrix $\mathbf{W}$.

6.1 Experimental Setup

We evaluate our methods in a dynamic setting where clients may opt out after 30% of the training rounds. When a client drops out, all its edges are removed, and the remaining clients continue to train collaboratively within the connected subnetwork.
We evaluate our methods on four image classification datasets: FashionMNIST [41], a 70000 sample grayscale clothing dataset; EMNIST Balanced [9], an extended handwritten character dataset built on MNIST; CIFAR 10 [25], a 60000 image benchmark with diverse objects; and CelebA [29], a facial attributes dataset with over 200000 celebrity images annotated with 40 binary attributes.

For FashionMNIST we use a two-layer MLP, for EMNIST we use LeNet-5, for CIFAR-10 we use a three-layer convolutional network, and for CelebA we use a small convolutional network with global average pooling; full architectural details are provided in Appendix E.

To model non-iid data across clients, we partition each dataset across nodes using a Dirichlet-based label partitioning scheme, following the approach in [1] with $\alpha = 0.1$ (including for the balanced datasets) while ensuring that each client receives data from the same number of datapoints. We construct each client's test distribution as a mixture: 60% mirrors the client's local training distribution, while the remaining 40% is drawn from a uniform label distribution. This setup lets us evaluate both performance on familiar patterns and adaptation to unseen or underrepresented labels.

Additional implementation details, including hyperparameters and client threshold selection are provided in Appendix E.

6.2 Experimental Results

Table 1 reports the preferred-model test accuracy (averaged over clients) and client retention rate for `NodeDrop-IDSGD` (Ours); decentralized SGD without dropouts (Base); and with voluntary dropouts (Drop.) across four datasets and four network topologies.

Allowing dropouts reduces accuracy and can severely reduce client retention, especially on CIFAR-10. `NodeDrop-IDSGD` substantially recovers this loss: it improves accuracy over the dropout baseline by up to 13 percentage points on FashionMNIST (SFL18), 33 percentage points on EMNIST (3Con16), 18 percentage points on CIFAR-10 (3Con16), and 14 percentage points on CelebA (3Con16); and increases retention with the largest gains on EMNIST.

The only different case is KCN, where our method yields negligible accuracy gains on FashionMNIST and CelebA and no improvement in retention for FashionMNIST, but still improves accuracy on EMNIST and CIFAR-10 (34.3% to 49.4%, 37.4% to 48.1%) and retention for EMNIST, CIFAR-10, and CelebA (50.0% to 75.9%; 0.0% to 36.2%; 3.8% to 20.3%).

These results show the gains from `NodeDrop-IDSGD` depend jointly on topology and data. On the scale-free graphs, voluntary dropouts hurt performance because dissatisfied hub clients both fragment the graph and remove rare labels from the collaboration. On 3Con16 even a few dropouts can disconnect the graph, so our method is especially effective at keeping clients in the collaboration, preserving connectivity, with the best results on EMNIST where many labels make non-IID effects particularly severe. In contrast, on the larger and more redundant KCN topology, where many nodes and multiple alternative paths make the graph less sensitive to individual dropouts FashionMNIST and CelebA already maintain strong performance under the dropout baseline, so incentivization adds little, whereas EMNIST still benefits because its higher label diversity and heterogeneity make them more sensitive to losing a small number of important clients. For CIFAR-10, the dropout baseline yields zero retention because of the combination of higher task complexity (compared with MNIST style datasets) and strongly non-IID client data , showing that clients prefer solo training and causing the collaboration to collapse entirely; in these cases, `NodeDrop-IDSGD` is the only way to sustain a connected component.

Table 2 compares the test accuracy of our solution under different strategies a_i across datasets and topologies. Hybrid rules achieve the best results (*Degree with KL* reaches 77.1% on FashionMNIST),

		Fashion MNIST				EMNIST				CIFAR-10				CelebA			
		Base	Drop.	**Ours**	Δ	Base	Drop.	**Ours**	Δ	Base	Drop.	**Ours**	Δ	Base	Drop.	**Ours**	Δ
SFL12	Accuracy	75.2	69.0	77.1	+8.10	57.1	46.8	70.5	+23.7	36.8	31.4	48.1	+16.7	61.7	61.1	72.4	+11.3
	Ret. Rate	N/A	44.2	81.2	+37.0	N/A	73.3	100	+26.7	N/A	0.0	21.7	+21.7	N/A	4.2	41.7	+37.5
3Con16	Accuracy	74.1	66.6	74.7	+8.10	54.1	36.3	69.2	**+32.9**	31.9	25.3	42.8	+17.5	58.7	57.0	70.8	+13.8
	Ret. Rate	N/A	37.5	64.4	+26.9	N/A	51.3	99.4	**+48.1**	N/A	0.00	11.3	+11.3	N/A	1.30	23.8	+22.5
SFL18	Accuracy	74.9	66.1	78.4	+12.3	57.2	43.6	66.2	+22.6	37.3	32.5	48.3	+15.8	60.7	60.3	70.4	+10.1
	Ret. Rate	N/A	41.7	65.0	+23.3	N/A	66.3	98.9	+32.6	N/A	0.00	27.8	+27.8	N/A	8.90	30.6	+21.7
KCN	Accuracy	77.0	68.2	69.1	+0.90	59.6	34.3	49.4	+15.1	41.9	37.4	48.1	+10.7	65.9	65.6	66.4	+0.80
	Ret. Rate	N/A	55.9	55.9	+0.00	N/A	50.0	75.9	+25.9	N/A	0.00	36.2	+36.2	N/A	3.80	20.3	+16.5

Table 1: Test accuracy (%) of the preferred model and client retention (%) in the largest connected component after dropouts for decentralized learning baselines (Base: no dropouts; Drop.: dropouts allowed) and the proposed incentivized method (Ours) across four datasets and four network topologies. Each configuration is evaluated over 10 random seeds, and results are averaged. Δ indicates the absolute improvement (percentage points) of our method over the baseline with dropouts allowed.

while data-based rules are most competitive on CelebA (*KL Divergence* attains 72.4% on SFL12 and 70.4% on SFL18). Results indicate that no strategy dominates; different strategies have indeed different impact (for FashionMNIST with SFL18, *Betweenness* reaches 53.1% while *Vertex connectivity* reachs 70%); and a given strategy can be best in one setting and among the worst in another (*Betweenness* for FashionMNIST with SFL18 versus CIFAR-10 withSFL12), underscoring the need to tailor a_i to both graph structure and data distribution.

7 Concluding Remarks

In this paper, we investigate the impact of client dropouts on the overall performance of decentralized learning systems and explore incentive mechanisms to mitigate these effects. We propose a generalized framework that incorporates dropout penalties directly into the objective function, enabling the system to discourage early exits. Our proposed solution, supported by convergence guarantees, show that the framework effectively incentivizes client participation, leading to retention gains of up to 48 percentage points and, consequently, better generalization and higher test accuracy, with accuracy gains of up to 33 percentage points in our experiments.

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	Fashion MNIST				EMNIST				CIFAR-10				CelebA			
Strategy (a_i)	SFL12	3Con16	SFL18	KCN	SFL12	3Con16	SFL18	KCN	SFL12	3Con16	SFL18	KCN	SFL12	3Con16	SFL18	KCN
Topology-based																
Degree	73.1	74.4	59.1	58.1	69.6	68.3	59.9	40.5	43.8	42.3	47.1	43.2	68.3	70.0	69.4	65.7
Degree (norm.)	73.7	73.6	58.6	65.3	67.8	66.7	58.7	46.2	43.3	41.8	45.6	44.0	69.6	69.8	69.5	65.8
Betweenness	67.9	74.4	53.1	68.3	68.2	68.2	63.5	40.8	**48.1**	42.3	44.9	46.8	63.2	70.0	62.7	64.2
Eigenvector	66.9	73.5	63.2	54.7	67.3	66.7	55.0	33.1	43.3	41.8	44.8	38.9	69.6	69.8	69.8	64.6
Closeness	76.8	70.8	62.2	58.2	70.2	**69.2**	65.6	33.8	43.2	42.5	45.4	40.7	70.3	70.0	69.3	63.8
PageRank	73.4	73.5	57.3	62.0	67.8	66.7	58.8	42.1	43.2	41.8	45.6	44.1	69.8	69.8	69.7	**66.4**
Vertex connect.	70.6	74.2	**70.0**	**69.1**	68.3	68.1	59.0	30.9	42.7	42.0	45.8	45.5	69.7	69.9	67.8	62.8
Data-based																
KL Divergence	72.3	74.7	61.8	60.4	69.3	68.2	56.6	40.5	42.7	42.1	43.9	44.9	**72.4**	70.6	**70.4**	62.6
Inverse Entropy	69.2	73.7	64.3	61.3	68.3	67.4	55.7	31.5	43.3	41.9	45.1	42.7	69.3	68.2	69.6	62.6
Dropout Risk	75.9	70.5	54.6	64.1	68.4	67.5	55.9	30.2	43.4	42.1	43.6	44.6	69.8	70.7	69.1	62.6
Hybrid																
Degree_x_KL	**77.1**	71.1	66.8	62.0	**70.5**	69.2	**66.2**	47.1	45.3	42.8	47.7	**48.1**	68.6	70.8	68.0	65.1
Betweenness_x_KL	69.9	71.1	60.7	61.2	68.6	69.1	63.1	41.8	46.0	**42.8**	45.3	46.7	63.1	**70.8**	56.7	63.4
Betweenness_x_DR	68.5	70.9	59.3	62.5	68.3	68.8	63.0	41.7	46.2	42.5	45.3	46.7	63.1	70.7	56.8	63.1
Eigenvector_x_KL	73.4	**74.7**	63.5	60.1	69.3	68.2	65.0	36.1	44.6	42.1	48.1	38.6	67.7	70.6	69.4	63.2
PageRank_x_KL	76.9	74.7	62.1	63.9	69.6	68.2	59.5	**49.4**	45.0	42.1	**48.3**	44.4	67.8	70.6	68.8	64.6

Table 2: Test accuracy (%) of `NodeDrop-IDSGD` under different importance weight strategies a_i across four datasets and four network topologies. Each configuration is evaluated over 10 random seeds, and results are averaged. For each dataset–topology column, the best-performing strategy is shown in bold and underlined.

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A Proposed Incentivized Learning Algorithm (Alternative)

We also propose an incentivized decentralized learning algorithm for the modified objective function in (8). Similar to Algorithm 2, it maintains the local model update procedure at the clients unchanged only adaptively adjusting the learning rate at the beginning of each round (lines 3-8) to perform this minimization.

A.1 Modified Objective with group-wise Penalty: `GroupDrop-IDSGD`

Algorithm 3 `GroupDrop-IDSGD`

Require: For each node $i \in [N]$, initialize $\mathbf{x}_i^{(0)} \in \mathbb{R}^d$, base LR $\{\eta\}_{t=0}^{T-1}$, no. of iterations T, mixing matrix $\mathbf{W}$

1: **for** $t \in 0 \dots T$ **do**
2: **For each worker** $i, i \in [N]$**:**
3: **Learning Rate Estimation Phase:**
4: Compute $\sigma(\mathbf{x}_i^{(t,0)})$
5: Compute M_{ki} $\forall k \in [K]$, $i \in [N]$ using log gossip (τ_b steps). `// low dim vector averaging`
6: Estimate $\hat{b}_i^{(t)} = (1-\gamma) + \gamma \sum_{k \in \mathcal{K}_i} \lambda_k M_{ki}^{(t)} (1 - \sigma(\mathbf{x}_i^{(t,0)}))$
7: Compute scaling factors $s_i^{(t)}$ using gossip averaging (τ_η steps). `// scalar averaging`
8: Set the effective learning rate $\hat{\eta}_i^{(t)} = \frac{\eta \hat{b}_i^{(t)}}{\tau(s_i^{(t)} + \epsilon)}$.
9: **Model Update Phase:**
10: **For** $r = 0, 1 \cdots (\tau - 1)$ **local iterations**
11: Sample $\xi_i^{(t,r)}$,
12: Compute $\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})$
13: $\mathbf{x}_i^{(t,r+1)} := \mathbf{x}_i^{(t,r)} - \hat{\eta}_i^{(t)} \nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})$ `// stochastic gradient updates`
14: $\mathbf{x}_i^{(t+1,0)} := \sum_{j \in \mathcal{N}_i} w_{ij} \mathbf{x}_j^{(t,\tau)}$ `// model gossip averaging`
15: **end for**

The gradient of the modified global objective function (equation (8)) is given by

$$\nabla G(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} (1-\gamma) \nabla F_i(\mathbf{x}) + \frac{1}{N} \gamma \sum_{k=1}^{K} \sum_{l \in C_k} \lambda_k \left(\prod_{j \in C_k} \sigma_j(\mathbf{x}) \right) (1 - \sigma_l(\mathbf{x})) \nabla F_l(\mathbf{x}). \tag{9}$$

Let $\mathcal{K}_i := \{k | i \in C_k\} \subseteq [K]$ be the set of all groups which node i is a part of. One can rewrite the gradient of the modified objective in equation (9) as

$$\nabla G(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} (1-\gamma) \nabla F_i(\mathbf{x}) + \frac{1}{N} \sum_{i=1}^{N} \left(\gamma \sum_{k \in \mathcal{K}_i} \lambda_k \left(\prod_{j \in C_k} \sigma_j(\mathbf{x}) \right) (1 - \sigma_i(\mathbf{x})) \right) \nabla F_i(\mathbf{x}).$$

Learning rate estimation phase: Note that this gradient can be expressed as a summation of terms over i. Every object in the ith term is computable at node i, except for $\left(\prod_{j \in C_k} \sigma_j(\mathbf{x}) \right)$. This product for $k \in \mathcal{K}_i$ must be estimated at node i in order to compute the gradient of the global objective function using the existing gossip infrastructure. Let $M_k^{(t)} := \prod_{j \in C_k} \sigma_j(\mathbf{x}_j^{(t,0)})$ denote the product associated with the kth group, and let $M_{ki}^{(t)}$ represent its estimate at node i at the beginning of round t. Note that this estimation is carried out only once every τ local SGD steps at each node. Once the estimation is completed, node i approximates the pre-factor $b_i^{(t)}$ at round t (line 6) as

$$\hat{b}_i^{(t)} := (1-\gamma) + \gamma \sum_{k \in \mathcal{K}_i} \lambda_k M_{ki}^{(t)} (1 - \sigma_i(\mathbf{x}_i^{(t,0)})).$$

Estimating Pre-factors: To estimate $M_k^{(t)} := \prod_{j \in C_k} \sigma_j(\mathbf{x}_j^{(t,0)})$, we use τ_b steps of log gossip. The idea of log gossip is simple – given a method that computes the average of N terms, one can estimate the product of N terms $[d_1, d_2, \dots, d_N]$ by first taking their logarithms and scaling by N to get $[N \ln(d_1), N \ln(d_2), \dots, N \ln(d_N)]$, averaging these via gossip over the network, and then exponentiating the result to

recover $\prod_i d_i$. Since the decentralized network topology serves as the communication backbone, every node participates in the log gossip procedure for all K groups. Define $\mathbf{m}_i^{(t,r_b)} := \left[M_{1i}^{(t,r_b)}, M_{2i}^{(t,r_b)} \cdots M_{Ki}^{(t,r_b)}\right]^\top \in \mathbb{R}^K$ as the vector of estimates at node i during the averaging step r_b at round t where $0 \le r_b \le \tau_b - 1$. Define the point wise log of this vector by

$$\begin{aligned}\mathbf{v}_i^{(t,r_b)} &\triangleq \ln\left(\mathbf{m}_i^{(t,r_b)}\right)\\ &= \left[\ln\left(M_{1i}^{(t,r_b)}\right), \ln\left(M_{2i}^{(t,r_b)}\right) \cdots \ln\left(M_{Ki}^{(t,r_b)}\right)\right]^\top\\ &= \left[v_{1i}^{(t,r_b)}, v_{2i}^{(t,r_b)} \cdots v_{Ki}^{(t,r_b)}\right]^\top.\end{aligned}$$

We perform gossip averaging on this vector. Specifically, we initialize

$$v_{ki}^{(t,0)} = \begin{cases} N \ln\left(\sigma_i(\mathbf{x}_i^{(t,0)})\right), & \text{if } k \in \mathcal{K}_i,\\ 0, & \text{otherwise.}\end{cases}$$

The update rule for gossip averaging follows the standard consensus form given by $\mathbf{v}_i^{(t,r_b+1)} = \sum_{j\in\mathcal{N}_i} w_{ij}\mathbf{v}_i^{(t,r_b)}$ or equivalently for the matrix $\mathbf{V}^{(t,r_b)} = \left[\mathbf{v}_1^{(t,r_b)}, \mathbf{v}_2^{(t,r_b)} \cdots \mathbf{v}_N^{(t,r_b)}\right] \in \mathbb{R}^{N\times K}$ the update rule in matrix form is given by $\mathbf{V}^{(t,r_b+1)} = \mathbf{V}^{(t,r_b)}\mathbf{W}$. After τ_b steps of log gossip we compute the final estimate $M_{ki}^{(t)}$ as

$$M_{ki}^{(t)} = \min\left\{M_{ki}^{(t,\tau_b)}, \left(M_{ki}^{(t,0)}\right)^{\frac{1}{N}}\right\}.$$

Where $M_{ki}^{(t,\tau_b)} = \exp(v_{ki}^{(t,\tau_b)})$ and $M_{ki}^{(t,0)} = \exp(v_{ki}^{(t,0)})$. Which is equivalent to

$$M_{ki}^{(t,\tau_b)} = \begin{cases} \min\left\{M_{ki}^{(t,\tau_b)}, \sigma_i(\mathbf{x}_i^{(t,0)})\right\} & k \in \mathcal{K}_i\\ \min\left\{M_{ki}^{(t,\tau_b)}, 1\right\} = M_{ki}^{(t,\tau_b)} & k \notin \mathcal{K}_i\end{cases}$$

The minimization is performed to ensure non-degenerate estimates, wherein, when $k \in \mathcal{K}_i$, the estimate $M_{k,i}^{(t)}$ of $M_k^{(t)} := \prod_{j\in\mathcal{C}_k} \sigma_j(\mathbf{x}_j^{(t,0)})$ does not exceed $\sigma_i(\mathbf{x}_i^{(t,0)})$ (since $\sigma_i(\cdot) \le 1$), thereby ensuring a reasonable approximation. Lastly, note that these τ_b steps of averaging are performed over vectors of dimension K, resulting in a negligible communication cost compared to the model averaging step, where nodes exchange high-dimensional models, which in the case of deep neural networks can potentially have hundreds of millions of parameters.

REMARK 4 (MINIMAL COMMUNICATION OVERHEAD FOR PRE-FACTOR AND SCALING FACTOR ESTIMATION). *The τ_η and τ_b steps of gossip averaging used for estimating the scaling factor (a one-dimensional quantity) and the pre-factor (products M_{ki} estimated jointly as K-dimensional vector) respectively, incur negligible communication cost as compared to the model averaging step, which involves exchanges of high-dimensional deep neural network parameters (line 14 on Algorithm 3).*

THEOREM 2 (CONVERGENCE OF GROUPDROP-IDGSD). *For the objective function G with groupwise penalty defined in equation* (8)*, under assumptions 1-4 there exists a base learning rate η under which for Algorithm 3 we have*

$$\begin{aligned}&\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G\left(\bar{\mathbf{x}}^{(t,0)}\right)\right\|^2\right] \le \mathcal{O}\left(\sqrt{\frac{(G_{\text{diff}}))\,(1+\epsilon_\eta)}{N\tau T}}\right)\\ &+ \mathcal{O}\left(\frac{(G_{\text{diff}})^{2/3}N^{1/3}}{p^{2/3}T^{2/3}}(1+\epsilon_\eta^{1/3})\right) + \mathcal{O}\left(\frac{(G_{\text{diff}})^{2/3}N^{1/3}}{p^{2/3}\tau^{1/3}T^{2/3}}(1+\epsilon_\eta)^{1/3}\right)\\ &+ \mathcal{O}\left(\frac{N\Xi_0^2}{pT}\right) + \mathcal{O}\left(\frac{(G_{\text{diff}})}{T}\right) + \mathcal{O}\left(\epsilon_\eta\right) + \mathcal{O}\left(\epsilon_b\right)\end{aligned}$$

Here $G_{\text{diff}} = G(\bar{\mathbf{x}}^{(0,0)}) - G_{\text{inf}}$,is the initial distance from the infimum and, $\mathcal{O}$ subsumes all constants. The parameter τ denotes the number of local steps performed in each iteration. The terms $\epsilon_\eta = \frac{N^2(1-p)^{\tau_\eta}}{4}$ and $\epsilon_b = (1-p)^{\tau_b}K_{\max}(N-1)\sum_{k=1}^K \lambda_k^2$, represent the learning rate estimation error and the pre-factor estimation error, respectively. Here, τ_η is the number of gossip averaging steps used for estimating the scaling factors, and τ_b is the number of steps of log *gossip used to estimate the product terms $M_{ki}^{(t)}$, $K_{\max} = \max_i |\mathcal{K}_i|$, the maximum number of groups a single node is the part of. The initial consensus error is defined as $\Xi_0^2 = \frac{1}{N}\sum_{i=1}^N \|\bar{\mathbf{x}}^{(0,0)} - \mathbf{x}_i^{(0,0)}\|^2$.*

See Appendix D for the proof. The terms that do not vanish as $T \to \infty$ are the learning rate estimation error $\epsilon_\eta = \frac{N^2(1-p)^{\tau_\eta}}{4}$ and pre-factor estimation error $\epsilon_b = (1-p)^{\tau_b}K_{\max}(N-1)\sum_{k=1}^K \lambda_k^2$. These terms, however, can be made arbitrarily small by increasing τ_η and τ_b, independent of the other terms with negligible communication overhead (see Remark 4). We observe a linear speedup in both the number of clients N and the number of local steps τ for the dominant term ($\mathcal{O}(T^{-1/2})$), which decays the slowest with the number of iterations T. There is a slowdown dependent on N in some of the faster decaying non-dominant terms, this happens because in the group wise penalty the gradient to be computed

at one client can potentially depend on the state of other clients. However for large values of T these terms decay fast and have less of an effect on convergence.

B Useful inequalities

PROPOSITION 3. *For a set of n vectors $\{\mathbf{w}_i\}_{i=1}^n$, where $\mathbf{w}_i \in \mathbb{R}^d$, the following inequality holds:*

$$\left\|\sum_{i=1}^n \mathbf{w}_i\right\|^2 \leq n \sum_{i=1}^n \|\mathbf{w}_i\|^2.$$

Moreover, for any two vectors $\mathbf{w}, \mathbf{v} \in \mathbb{R}^d$, we have:

$$\|\mathbf{w}+\mathbf{v}\|^2 \leq (1+\alpha)\|\mathbf{w}\|^2 + (1+\alpha^{-1})\|\mathbf{v}\|^2, \quad \forall \alpha > 0.$$

This inequality also extends to matrices $W, V \in \mathbb{R}^{n\times d}$ in the Frobenius norm.

PROPOSITION 4 (ADAPTED VERSION OF LEMMA 4 IN [22]). *Let $\{X_i\}_{i=1}^N$ be N random variables in $\mathbb{R}^d$, which may not necessarily be independent. Suppose that the conditional expectation satisfies $\mathbb{E}[X_i|X_{i-1},\ldots,X_1] = Y_i$, and that the variance condition $\mathbb{E}[\|X_i - Y_i\|_2^2] \leq \sigma_i^2$ holds. Then, we have*

$$\mathbb{E}\left[\left\|\sum_{i=1}^N (X_i - Y_i)\right\|_2^2\right] \leq \sum_{i=1}^N \sigma_i^2.$$

This result is used while bounding the variance of local gradient updates.

LEMMA 1 (TUNING THE LEARNING RATE - LEMMA 17 IN [23]). *For any non-negative parameters A_0, A_2, A_3, A_4, there exists a constant learning rate η satisfying $\eta \leq \frac{1}{A_4}$ such that*

$$\Psi_T = \frac{A_0}{\eta T} + A_2\eta + A_3\eta^2 \leq 2\left(\frac{A_2 A_0}{T}\right)^{\frac{1}{2}} + 2A_3^{\frac{1}{3}}\left(\frac{A_0}{T}\right)^{\frac{2}{3}} + \frac{A_4 A_0}{T}.$$

The learning rate is chosen as $\eta = \min\left\{\left(\frac{A_0}{A_2 T}\right)^{\frac{1}{2}}, \left(\frac{A_0}{A_3 T}\right)^{\frac{1}{3}}, \frac{1}{A_4}\right\} \leq \frac{1}{A_4}$.

C Modified Objective with Node-wise Penalty

In this section, we define the following terms, which will be used exclusively within this context.

$$\mathbf{X}^{(t,0)} := \left[\mathbf{x}_1^{(t,0)}, \ldots, \mathbf{x}_N^{(t,0)}\right] \in \mathbb{R}^{d\times N},$$

$$\bar{\mathbf{X}}^{(\mathbf{t},\mathbf{0})} := \left[\bar{\mathbf{x}}^{(t,0)}, \ldots, \bar{\mathbf{x}}^{(t,0)}\right] = \mathbf{X}^{(t,0)} \cdot \frac{1}{N}\mathbf{1}\mathbf{1}^\top \in \mathbb{R}^{d\times N}$$

where $\bar{\mathbf{x}}^{(t,0)} = \frac{1}{N}\sum_{i=1}^N \mathbf{x}_i^{(t,0)}$. The modified objective function with node wise penalty is defined as follows

$$G(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N G_i(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N ((1-\gamma)F_i(\mathbf{x}) + \gamma a_i \sigma_i(\mathbf{x})) \tag{10}$$

and its gradient at $\mathbf{x}$ is given by $\nabla G(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N \nabla G_i(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N ((1-\gamma) + \gamma a_i \sigma_i(\mathbf{x})(1-\sigma_i(\mathbf{x})))\nabla F_i(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N ((1-\gamma) + \gamma a_i q_i(\mathbf{x}))\nabla F_i(\mathbf{x})$ where $q_i(\mathbf{x}) = \sigma_i(\mathbf{x})(1-\sigma_i(\mathbf{x}))$. We also define the aggregated update at client i and its noiseless value as

$$\mathbf{h}_i^t = \left((1-\gamma) + \gamma q_i(\mathbf{x}_i^{(t,0)})a_i\right)\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)}) = b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)}) \tag{11}$$

$$\bar{\mathbf{h}}_i^t = \left((1-\gamma) + \gamma q_i(\mathbf{x}_i^{(t,0)})a_i\right)\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)}) = b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)}). \tag{12}$$

Here $b_i^{(t)} = (1-\gamma) + \gamma q_i(\mathbf{x}_i^{(t,0)})a_i$. Note that $\nabla G_i(\mathbf{x}_i^{(t,0)}) = \left((1-\gamma) + \gamma q_i(\mathbf{x}_i^{(t,0)})a_i\right)\nabla F_i(\mathbf{x}_i^{(t,0)}) = \nabla b_i^{(t)} F_i(\mathbf{x}_i^{(t,0)})$. Define $\tilde{\eta}^{(t)} := \frac{\eta}{\bar{s}^{(t)}+\epsilon}$ which is the effective learning rate w.r.t. the modified objective G at $(t,0)$ and $\tilde{\eta}_i^{(t)} := \frac{\eta}{s_i^{(t)}+\epsilon}$ is the estimate of $\tilde{\eta}^{(t)}$ at node $i \in [N]$. Note that the update at node i during round r can be rewritten as

$$\mathbf{x}_i^{(t,r+1)} := \mathbf{x}_i^{(t,r)} - \hat{\eta}_i^{(t)}\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)}) = \mathbf{x}_i^{(t,r)} - \tilde{\eta}_i^{(t)} b_i^{(t)}\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})/\tau. \tag{13}$$

For $a_{\max} = \max_i a_i$ we define the constant

$$w_{\max} = 4(1-\gamma) + a_{\max}\gamma. \tag{14}$$

We shall repeatedly used the inequality $(1-\gamma)+\gamma a_i q_i(\mathbf{x}) \leq (1-\gamma)+\gamma a_{\max} q_i(\mathbf{x}) \leq (1-\gamma)+\frac{\gamma a_{\max}}{4} = \frac{w_{\max}}{4}$. We also have the pre-factor $b_i^{(t)} \leq \frac{w_{\max}}{4}\ \forall i, t$. From our update rule and $\tilde{L}_s$ smoothness of function G from lemma 2 we have

$$\mathbb{E}\left[G(\bar{\mathbf{x}}^{(t+1,0)})\right] - \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t,0)})\right] \leq -\underbrace{\mathbb{E}\left[\left\langle \nabla G(\bar{\mathbf{x}}^{(t,0)}), \left(\frac{1}{\tau N}\sum_{i=1}^{N}\tilde{\eta}_i^{(t)}\mathbf{h}_i^t\right)\right\rangle\right]}_{T_1} + \underbrace{\mathbb{E}\left[\frac{\tilde{L}_s}{2}\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\tilde{\eta}_i^{(t)}\mathbf{h}_i^t\right)\right\|^2\right]}_{T_2}. \tag{15}$$

We first focus on simplifying the term T_2.

$$T_2 = \mathbb{E}\left[\frac{\tilde{L}_s}{2}\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\tilde{\eta}_i^{(t)}\mathbf{h}_i^t\right)\right\|^2\right] \tag{16}$$

$$= \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2}{2}\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\mathbf{h}_i^t\right)\right\|^2\right] \tag{17}$$

$$\leq \tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\bar{\mathbf{h}}_i^t\right)\right\|^2\right] + \underbrace{\tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\left(\mathbf{h}_i^t - \bar{\mathbf{h}}_i^t\right)\right)\right\|^2\right]}_{T_3} \tag{18}$$

T_3 here can be bounded as

$$T_3 = \tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\left(\mathbf{h}_i^t - \bar{\mathbf{h}}_i^t\right)\right)\right\|^2\right] \tag{19}$$

$$= \frac{\tilde{L}_s}{\tau^2 N^2}\mathbb{E}\left[\sum_{i=1}^{N}(\tilde{\eta}_i^{(t)})^2\left\|\left(b_i^{(t)}\sum_{r=0}^{\tau-1}\left(\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)}) - F_i(\mathbf{x}_i^{(t,r)})\right)\right)\right\|^2\right] \tag{20}$$

$$\leq \frac{\tilde{L}_s}{\tau N^2}\sum_{i=1}^{N}(\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2\sigma^2 \tag{21}$$

$$\leq \frac{2\tilde{L}_s}{\tau N^2}\sum_{i=1}^{N}(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\sigma^2 + \frac{2\tilde{L}_s}{\tau N^2}\sum_{i=1}^{N}\|\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)}\|^2(b_i^{(t)})^2\sigma^2 \tag{22}$$

$$\leq \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau N} + \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2\epsilon_\eta}{8\tau N}. \tag{23}$$

Here equation (21) follows from proposition 4 and equation 20 follows from independent sampling of mini-batches by the clients, (23) follows from the fact $\tilde{\eta}^{(t)}b_i^{(t)} \leq \eta$, $b_i^{(t)} \leq \frac{w_{\max}}{4}$ and lemma 3. Thus we have

$$T_2 \leq \tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\bar{\mathbf{h}}_i^t\right)\right\|^2\right] + \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau N}(1+\epsilon_\eta). \tag{24}$$

Consider the term T_1

$$T_1 = -\mathbb{E}\left[\left\langle \nabla G(\bar{\mathbf{x}}^{(t,0)}), \left(\frac{1}{\tau N}\sum_{i=1}^{N}\tilde{\eta}_i^{(t)}\bar{\mathbf{h}}_i^t\right)\right\rangle\right] - \mathbb{E}\left[\left\langle \nabla G(\bar{\mathbf{x}}^{(t,0)}), \left(\frac{1}{\tau N}\sum_{i=1}^{N}\tilde{\eta}_i^{(t)}(\mathbf{h}_i^t - \bar{\mathbf{h}}_i^t)\right)\right\rangle\right] \tag{25}$$

$$= -\frac{\tilde{\eta}^{(t)}}{2}\left\|\nabla G(\bar{\mathbf{x}}^{(t,0)})\right\|^2 - \underbrace{\frac{\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\bar{\mathbf{h}}_i^t\right\|^2}_{T_4} + \underbrace{\frac{\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\nabla G(\bar{\mathbf{x}}^{(t,0)}) - \frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\bar{\mathbf{h}}_i^t\right\|^2}_{T_5}. \tag{26}$$

We can bound T_5 as follows

$$T_5 = \frac{\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\nabla G(\bar{\mathbf{x}}^{(t,0)}) - \frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\bar{\mathbf{h}}_i^t\right\|^2 \tag{27}$$

$$\leq \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\nabla G(\bar{\mathbf{x}}^{(t,0)}) - \frac{1}{N}\sum_{i=1}^{N}\nabla G_i(\mathbf{x}_i^{(t,0)})\right\|^2 + \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N}\sum_{i=1}^{N}\nabla G_i(\mathbf{x}_i^{(t,0)}) - \frac{1}{\tau N}\sum_{i=1}^{N}\bar{\mathbf{h}}_i^t\right\|^2 \tag{28}$$

$$+ \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\left(1-\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\right)\right\|^2$$

$$\leq \frac{3\tilde{\eta}^{(t)}}{2N}\sum_{i=1}^{N}\mathbb{E}\left\|\nabla G_i(\bar{\mathbf{x}}^{(t,0)})-\nabla G_i(\mathbf{x}_i^{(t,0)})\right\|^2 + \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N}\sum_{i=1}^{N}\nabla G_i(\mathbf{x}_i^{(t,0)})-\frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\right\|^2 \tag{29}$$

$$+ \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{\tau N}\sum_{i=1}^{N}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F(\mathbf{x}_i^{(t,r)})\left(1-\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\right)\right\|^2$$

$$\leq \frac{3\tilde{\eta}^{(t)}\tilde{L}_s^2}{2N}\sum_{i=1}^{N}\mathbb{E}\left\|\bar{\mathbf{x}}^{(t,0)}-\mathbf{x}_i^{(t,0)}\right\|^2 + \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N}\sum_{i=1}^{N}\nabla G_i(\mathbf{x}_i^{(t,0)})-\frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\right\|^2 \tag{30}$$

$$+ \frac{3\tilde{\eta}^{(t)}L_c^2}{2N}\sum_{i=1}^{N}\frac{\|\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\|^2}{\|\tilde{\eta}^{(t)}\|^2}(b_i^{(t)})^2$$

$$\leq \frac{3\tilde{L}_s^2\tilde{\eta}^{(t)}}{2N}\sum_{i=1}^{N}\mathbb{E}\left\|\bar{\mathbf{x}}^{(t,0)}-\mathbf{x}_i^{(t,0)}\right\|^2 + \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N}\sum_{i=1}^{N}\nabla G_i(\mathbf{x}_i^{(t,0)})-\frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\right\|^2 \tag{31}$$

$$+ \frac{3}{2}L_c^2\tilde{\eta}^{(t)}\max_i(b_i^{(t)})^2\epsilon_\eta$$

$$\leq \underbrace{\frac{3\tilde{L}_s^2\tilde{\eta}^{(t)}}{2N}\sum_{i=1}^{N}\mathbb{E}\left\|\bar{\mathbf{x}}^{(t,0)}-\mathbf{x}_i^{(t,0)}\right\|^2}_{\text{Consensus Error}} + \underbrace{\frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N}\sum_{i=1}^{N}\nabla G_i(\mathbf{x}_i^{(t,0)})-\frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\right\|^2}_{T_6} + \frac{3L_c^2\tilde{\eta}^{(t)}w_{\max}^2\epsilon_\eta}{32}. \tag{32}$$

Here equation (31) follows from lemma 4 and equation (32) follows from $\tilde{\eta}^{(t)}b_i^{(t)} \leq \eta$ and $b_i^{(t)} \leq \frac{w_{\max}}{4}$. We can bound T_6 as follows

$$T_6 = \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N}\sum_{i=1}^{N}\left(\nabla G_i(\mathbf{x}_i^{(t,0)})-\frac{1}{\tau}\overline{\mathbf{h}}_i^t\right)\right\|^2 \tag{33}$$

$$= \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N}\sum_{i=1}^{N}b_i^{(t)}\nabla F_i(\mathbf{x}_i^{(t,0)})-\frac{1}{\tau N}\sum_{i=1}^{N}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)})\right\|^2 \tag{34}$$

$$= \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N\tau}\sum_{i=1}^{N}b_i^{(t)}\sum_{r=0}^{\tau-1}\left(\nabla F_i(\mathbf{x}_i^{(t,0)})-\nabla F_i(\mathbf{x}_i^{(t,r)})\right)\right\|^2 \tag{35}$$

$$\leq \frac{3\tilde{\eta}^{(t)}}{2N\tau}\sum_{i=1}^{N}(b_i^{(t)})^2\sum_{r=0}^{\tau-1}\mathbb{E}\left[\left\|\nabla F_i(\mathbf{x}_i^{(t,0)})-\nabla F_i(\mathbf{x}_i^{(t,r)})\right\|^2\right] \tag{36}$$

$$\leq \frac{3\tilde{\eta}^{(t)}L_s^2}{2N\tau}\sum_{i=1}^{N}(b_i^{(t)})^2\sum_{r=0}^{\tau-1}\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] \tag{37}$$

$$\leq \frac{3\tilde{\eta}^{(t)}L_s^2}{2N\tau}\sum_{i=1}^{N}(b_i^{(t)})^2\Bigg(4(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\tau\mathbb{E}\left[\left\|\nabla F_i(\mathbf{x}_i^{(t,0)})\right\|^2\right] + 2\tau L_c^2\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]$$

$$+ 6\mathbb{E}\left[(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\sigma^2\right] + 6\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2\sigma^2\right]\Bigg) \tag{38}$$

$$\leq \frac{3\tilde{\eta}^{(t)}L_s^2}{2N\tau}\sum_{i=1}^{N}\Bigg(\frac{(\tilde{\eta}^{(t)})^2w_{\max}^2(b_i^{(t)})^2\tau\mathbb{E}\left[\left\|\nabla F_i(\mathbf{x}_i^{(t,0)})\right\|^2\right]}{4}$$

$$+\frac{w_{\max}^2\tau(L_c^2)\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]}{8} + \frac{3w_{\max}^4(\tilde{\eta}^{(t)})^2\sigma^2}{128} + \frac{3w_{\max}^2(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2\sigma^2}{8}\Bigg) \tag{39}$$

$$\leq \frac{3\tilde{\eta}^{(t)}L_s^2}{2N\tau}\sum_{i=1}^{N}\left(\frac{(\tilde{\eta}^{(t)})^2w_{\max}^2\tau\mathbb{E}\left[\left\|\nabla G_i(\mathbf{x}_i^{(t,0)})\right\|^2\right]}{4}\right) + \frac{3\tilde{\eta}^{(t)}L_s^2}{256N\tau}\left(w_{\max}^4(\tilde{\eta}^{(t)})^2\tau L_c^2N\epsilon_\eta + 3Nw_{\max}^4(\tilde{\eta}^{(t)})^2\sigma^2(1+\epsilon_\eta)\right) \tag{40}$$

$$= \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2}{8N} \sum_{i=1}^{N} \mathbb{E}\left[\left\|\nabla G_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right] + \frac{3(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 L_c^2}{256}\epsilon_\eta + \frac{9(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 \sigma^2}{256\tau}(1+\epsilon_\eta) \tag{41}$$

$$\leq \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2}{4N} \sum_{i=1}^{N} \mathbb{E}\left[\left\|\nabla G_i\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right] + \frac{3(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 L_c^2}{256}\epsilon_\eta + \frac{9(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 \sigma^2}{256\tau}(1+\epsilon_\eta)$$

$$+ \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2}{4N} \sum_{i=1}^{N} \mathbb{E}\left[\left\|\nabla G_i\left(\overline{\mathbf{x}}^{(t,0)}\right) - \nabla G_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right] \tag{42}$$

$$\leq \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2}{4N} \sum_{i=1}^{N} \mathbb{E}\left[\left\|\nabla G_i\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right] + \frac{3(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 L_c^2}{256}\epsilon_\eta + \frac{9(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 \sigma^2}{256\tau}(1+\epsilon_\eta)$$

$$+ \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2 \tilde{L}_s^2}{4N} \sum_{i=1}^{N} \mathbb{E}\left[\left\|\overline{\mathbf{x}}^{(t,0)} - \mathbf{x}_i^{(t,0)}\right\|^2\right] \tag{43}$$

$$\leq \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2 \tilde{\beta}^2}{4} \mathbb{E}\left[\left\|\nabla G\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right] + \frac{3(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 L_c^2}{256}\epsilon_\eta + \frac{9(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 \sigma^2}{256\tau}(1+\epsilon_\eta)$$

$$+ \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2 \tilde{L}_s^2}{4N} \sum_{i=1}^{N} \mathbb{E}\left[\left\|\overline{\mathbf{x}}^{(t,0)} - \mathbf{x}_i^{(t,0)}\right\|^2\right] + \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2 \tilde{\kappa}^2}{4}. \tag{44}$$

Here (39) follows from $b_i^{(t)} \leq \frac{w_{\max}}{4}$ and equation (38) follows from lemma 5. Equation (40) follows from lemma 3 and the definition of $G(\mathbf{x})$. Now from the equations (24), (26), (31) and (44) for $\Xi_t^2 := \frac{1}{N}\sum_{i=1}^{N} E\left[\left\|\overline{\mathbf{x}}^{(t,0)} - \mathbf{x}_i^{(t,0)}\right\|^2\right]$ we have

$$\mathbb{E}\left[G(\bar{\mathbf{x}}^{(t+1,0)})\right] - \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t,0)})\right] \leq -\frac{\tilde{\eta}^{(t)}}{2}\left\|\nabla G\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2 + \tilde{\eta}^{(t)}\left(\tilde{L}_s\tilde{\eta}^{(t)} - \frac{1}{2}\right)\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\overline{\mathbf{h}}_i^t\right)\right\|^2\right]$$

$$+ \left(\frac{3\tilde{L}_s^2\tilde{\eta}^{(t)}}{2} + \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2 \tilde{L}_s^2}{4}\right)\Xi_t^2 + \left(\frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau N} + \frac{9(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2}{256\tau}\right)\sigma^2$$

$$+ \frac{3(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^2 \tilde{\kappa}^2}{4} + \left(\frac{3L_c^2\tilde{\eta}^{(t)} w_{\max}^2\left(8 + (\tilde{\eta}^{(t)})^2 w_{\max}^2 L_s^2\right)}{256} + \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau N} + \frac{9(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2\sigma^2}{256\tau}\right)\epsilon_\eta$$

$$+ \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2 \tilde{\beta}^2}{4}\mathbb{E}\left[\left\|\nabla G\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right]. \tag{45}$$

Using $\tilde{\eta}^{(t)} \leq \min\left\{\frac{1}{4\tilde{L}_s}, \frac{\sqrt{2}}{\sqrt{3}L_s}\right\}$ we can simplify further

$$\tilde{\eta}^{(t)}\left(\frac{1}{2} - \frac{3L_s^2(\tilde{\eta}^{(t)})^2 w_{\max}^2\tilde{\beta}^2}{4}\right)\mathbb{E}\left[\left\|\nabla G\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right] \leq \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t,0)})\right] - \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t+1,0)})\right] + 2\tilde{L}_s^2\tilde{\eta}^{(t)}\Xi_t^2$$

$$+\tilde{\eta}^{(t)}\frac{3L_c^2 w_{\max}^2}{32}\epsilon_\eta + (\tilde{\eta}^{(t)})^2\left(\frac{\tilde{L}_s w_{\max}^2\sigma^2(1+\epsilon_\eta)}{8\tau N}\right) + (\tilde{\eta}^{(t)})^3\left(\frac{3L_c^2 L_s^2 w_{\max}^4}{256}\epsilon_\eta + \frac{9L_s^2 w_{\max}^4\sigma^2}{256\tau}(1+\epsilon_\eta) + \frac{3L_s^2 w_{\max}^2\tilde{\kappa}^2}{4}\right). \tag{46}$$

Using the fact $\tilde{\eta}^{(t)} \leq \frac{\eta}{\epsilon}$ and averaging over T we have

$$\frac{1}{T}\sum_{t=1}^{T}\left(\tilde{\eta}^{(t)}\left(\frac{1}{2} - \frac{3L_s^2(\tilde{\eta}^{(t)})^2 w_{\max}^2\tilde{\beta}^2}{4}\right)\mathbb{E}\left[\left\|\nabla G\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right]\right) \leq \frac{\sum_{t=1}^{T}\mathbb{E}\left[G(\bar{\mathbf{x}}^{(t,0)})\right] - \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t+1,0)})\right]}{T} + \frac{2\tilde{L}_s^2\eta}{\epsilon T}\sum_{t=1}^{T}\Xi_t^2$$

$$+\frac{3\eta L_c^2 w_{\max}^2}{32\epsilon}\epsilon_\eta + (\eta)^2\left(\frac{\tilde{L}_s w_{\max}^2\sigma^2(1+\epsilon_\eta)}{8\tau N\epsilon^2}\right) + (\eta)^3\left(\frac{3L_c^2 L_s^2 w_{\max}^4}{256\epsilon^3}\epsilon_\eta + \frac{9L_s^2 w_{\max}^4}{256\epsilon^3\tau\sigma^2}(1+\epsilon_\eta) + \frac{3L_s^2 w_{\max}^2\tilde{\kappa}^2}{4\epsilon^3}\right). \tag{47}$$

Now using lemma 6 we have

$$\frac{1}{T}\sum_{t=1}^{T}\tilde{\eta}^{(t)}\left[\frac{1}{2} - \tilde{\beta}^2\left(\frac{3L_s^2(\tilde{\eta}^{(t)})^2 w_{\max}^2}{4} + \frac{256\tilde{L}_s^2\eta}{p^2\epsilon}\tilde{\eta}^{(t)}\right)\right]\mathbb{E}\left[\|\nabla G(\overline{\mathbf{x}}^{(t,0)})\|^2\right] \leq \frac{G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}}{T}$$

$$+\frac{2\tilde{L}_s^2\eta}{\epsilon}\left\{\frac{4\Xi_0^2}{pT}+\frac{128}{p^2T}\sum_{t=1}^{T}(\tilde{\eta}^{(t)})^2\,\tilde{\kappa}^2+\frac{2w_{\max}^2}{p^2T}\sum_{t=1}^{T}(\tilde{\eta}^{(t)})^2\,L_c^2\,\epsilon_n+\frac{3w_{\max}^2}{p^2\tau T}\sum_{t=1}^{T}(\tilde{\eta}^{(t)})^2\,\sigma^2(1+\epsilon_\eta)\right\}$$

$$+\frac{3\eta L_c^2 w_{\max}^2}{32\epsilon}\epsilon_\eta+(\eta)^2\Big(\frac{\tilde{L}_s w_{\max}^2\sigma^2(1+\epsilon_\eta)}{8\tau N\epsilon^2}\Big)+(\eta)^3\Big(\frac{3L_c^2L_s^2w_{\max}^4}{256\epsilon^3}\epsilon_\eta+\frac{9L_s^2w_{\max}^4\sigma^2}{256\epsilon^3\tau}(1+\epsilon_\eta)+\frac{3L_s^2w_{\max}^2\tilde{\kappa}^2}{4\epsilon^3}\Big). \tag{48}$$

Again using the fact $\frac{\eta}{\epsilon+\frac{Nw_{\max}}{4}}\le\tilde{\eta}^{(t)}\le\frac{\eta}{\epsilon}$ for $\eta\le\frac{\epsilon}{2\tilde{\beta}\sqrt{\left(\frac{L_s^2w_{\max}^2}{4}+\frac{256\tilde{L}_s^2}{p^2}\right)}}$ we have $\frac{1}{2}-\tilde{\beta}^2\left(\frac{L_s^2(\tilde{\eta}^{(t)})^2w_{\max}^2}{4}+\frac{256\tilde{L}_s^2\eta}{p^2\epsilon}\tilde{\eta}^{(t)}\right)\le\frac{1}{4}$ and the LHS can be lower bounded by $\frac{\eta}{4(\epsilon+Nw_{\max}/4)}\left(\frac{1}{T}\sum_{t=0}^{T-1}\left(\mathbb{E}\left[\left\|\nabla G\big(\overline{\mathbf{x}}^{(t,0)}\big)\right\|^2\right]\right)\right)$. Further substituting $\tilde{\eta}^{(t)}\le\frac{\eta}{\epsilon}$ in the RHS we have

$$\frac{\eta}{4(\epsilon+Nw_{\max}/4)}\left(\frac{1}{T}\sum_{t=0}^{T-1}\left(\mathbb{E}\left[\left\|\nabla G\big(\overline{\mathbf{x}}^{(t,0)}\big)\right\|^2\right]\right)\right)\le\frac{G(\bar{\mathbf{x}}^{(0,0)})-G_{\inf}}{T}$$

$$+\left(\frac{8\tilde{L}_s^2\,\Xi_0^2}{pT\,\epsilon}+\frac{3L_c^2w_{\max}^2}{32\epsilon}\epsilon_\eta\right)\eta+\left(\frac{\tilde{L}_s w_{\max}^2\sigma^2(1+\epsilon_\eta)}{8\tau N\,\epsilon^2}\right)\eta^2+$$

$$\left(\frac{256\tilde{L}_s^2\tilde{\kappa}^2}{p^2\,\epsilon^3}+\frac{4\tilde{L}_s^2w_{\max}^2L_c^2\,\epsilon_n}{p^2\,\epsilon^3}+\frac{6\tilde{L}_s^2w_{\max}^2\sigma^2}{p^2\tau\,\epsilon^3}(1+\epsilon_\eta)+\frac{3L_c^2L_s^2w_{\max}^4}{256\epsilon^3}\epsilon_\eta+\frac{9L_s^2w_{\max}^4\sigma^2}{256\epsilon^3\tau}(1+\epsilon_\eta)+\frac{3L_s^2w_{\max}^2\tilde{\kappa}^2}{4\epsilon^3}\right)\eta^3. \tag{49}$$

Choosing $\epsilon=\frac{N\gamma a_{\max}L_c^2}{6\sqrt{3}L_s}$ (see lemma 2) for making the step size $\tilde{\eta}^{(t)}$ inversely proportional to the smoothness parameter of the global objective G we can rewrite equation 49 as

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G\big(\overline{\mathbf{x}}^{(t,0)}\big)\right\|^2\right]\le\frac{A_0}{\eta T}+A_1+A_2\eta+A_3\eta^2 \tag{50}$$

where

$$A_0=\frac{N\,(2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s)}{3\sqrt{3}\,L_s}\Big[G(\bar{\mathbf{x}}^{(0,0)})-G_{\inf}\Big], \tag{51}$$

$$A_1=\frac{16\,(2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s)\,\tilde{L}_s^2\,\Xi_0^2}{pT\,\gamma\,a_{\max}L_c^2}+\frac{3\,(2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s)\,w_{\max}^2}{16\,\gamma\,a_{\max}}\,\epsilon_\eta, \tag{52}$$

$$A_2=\frac{9\,\tilde{L}_s\,w_{\max}^2\,\sigma^2\,(1+\epsilon_\eta)\,L_s\,(2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s)}{2\sqrt{3}\,\tau\,N^2\,\gamma^2a_{\max}^2L_c^4}, \tag{53}$$

$$A_3=\frac{216\,L_s^2\,(2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s)}{N^2\,\gamma^3a_{\max}^3L_c^6}\left[\frac{256\,\tilde{L}_s^2\,\tilde{\kappa}^2}{p^2}+\frac{4\,\tilde{L}_s^2\,w_{\max}^2\,L_c^2\,\epsilon_n}{p^2}+\frac{6\,\tilde{L}_s^2\,w_{\max}^2\,\sigma^2}{p^2\tau}(1+\epsilon_\eta)\right.$$

$$\left.+\frac{3\,L_c^2\,L_s^2\,w_{\max}^4}{256}\,\epsilon_\eta+\frac{9\,L_s^2\,w_{\max}^4\,\sigma^2}{256\tau}(1+\epsilon_\eta)+\frac{3\,L_s^2\,w_{\max}^2\,\tilde{\kappa}^2}{4}\right]. \tag{54}$$

For small values of gamma we can use the fact $\tilde{\eta}^{(t)}\le\frac{\eta}{N(1-\gamma)}$ and using the same steps as above starting from equation (45) we obtain the bound

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G\big(\overline{\mathbf{x}}^{(t,0)}\big)\right\|^2\right]\le\frac{A_0}{\eta T}+A_1+A_2\eta+A_3\eta^2 \tag{55}$$

for

$$A_0=\frac{N\Big(2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s\Big)}{3\sqrt{3}\,L_s}\Big[G\big(\bar{\mathbf{x}}^{(0,0)}\big)-G_{\inf}\Big], \tag{56}$$

$$A_1=\frac{2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s}{3\sqrt{3}\,L_s}\left[\frac{8\,\tilde{L}_s^2\,\Xi_0^2}{pT\,(1-\gamma)}+\frac{3L_c^2\,w_{\max}^2}{32\,(1-\gamma)}\,\epsilon_\eta\right], \tag{57}$$

$$A_2=\frac{(2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s)\,\tilde{L}_s\,w_{\max}^2\,\sigma^2\,(1+\epsilon_\eta)}{24\sqrt{3}\,L_s\,\tau\,N^2\,(1-\gamma)^2}, \tag{58}$$

$$A_3 = \frac{2\gamma\, a_{\max} L_c^2 + 3\sqrt{3}\, w_{\max} L_s}{3\sqrt{3}\, L_s N^2\,(1-\gamma)^3}\Bigg[\frac{256\,\tilde{L}_s^2\,\tilde{\kappa}^2}{p^2} + \frac{4\,\tilde{L}_s^2\, w_{\max}^2\, L_c^2\,\epsilon_n}{p^2} + \frac{6\,\tilde{L}_s^2\, w_{\max}^2\,\sigma^2}{p^2\tau}(1+\epsilon_\eta) + \frac{3L_c^2\, L_s^2\, w_{\max}^4}{256}\,\epsilon_\eta + \frac{9L_s^2\, w_{\max}^4\,\sigma^2}{256\tau}(1+\epsilon_\eta) + \frac{3L_s^2\, w_{\max}^2\,\tilde{\kappa}^2}{4}\Bigg]. \tag{59}$$

The final bound is obtained using lemma 1, taking the case wise minimum of the two expressions and choosing $A_4 = \max\left\{\frac{2\tilde{\beta}\sqrt{\left(\frac{L_s^2 w_{\max}^2}{4} + \frac{256\tilde{L}_s^2}{p^2}\right)}}{\epsilon}, \frac{4\tilde{L}_s}{\epsilon}, \frac{3}{\sqrt{2}\epsilon}\right\}$, we have for $\epsilon = \frac{N\gamma a_{\max} L_c^2}{6\sqrt{3}L_s}$ chosen as per lemma (2) to make the step size $\tilde{\eta}^{(t)}$ with respect to the modified objective G inversely proportional to the smoothness parameter

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G\big(\bar{\mathbf{x}}^{(t,0)}\big)\right\|^2\right] \le A_1 + 2\left(\frac{A_2 A_0}{T}\right)^{\frac{1}{2}} + 2A_3^{\frac{1}{3}}\left(\frac{A_0}{T}\right)^{\frac{2}{3}} + \frac{A_4 A_0}{T} \tag{60}$$

$$\begin{aligned}
&= \mathcal{O}\left(\epsilon_\eta\right) + \mathcal{O}\left(\sqrt{\frac{\big(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\big)\,(1+\epsilon_\eta)}{N\tau T}}\right) \\
&+ \mathcal{O}\left(\frac{\Xi_0^2}{pT}\right) + \mathcal{O}\left(\frac{(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf})}{T}\right) \\
&+ \mathcal{O}\Big(\frac{\big(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\big)^{2/3}}{T^{2/3}}\epsilon_\eta^{1/3}\Big) + \mathcal{O}\Big(\frac{\big(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\big)^{2/3}}{\tau^{1/3}T^{2/3}}(1+\epsilon_\eta)^{1/3}\Big) \\
&+ \mathcal{O}\Big(\frac{\big(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\big)^{2/3}}{p^{2/3}T^{2/3}}\Big) + \mathcal{O}\Big(\frac{\big(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\big)^{2/3}}{p^{2/3}T^{2/3}}\epsilon_\eta^{1/3}\Big) \\
&+ \mathcal{O}\Big(\frac{\big(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\big)^{2/3}}{p^{2/3}\tau^{1/3}T^{2/3}}(1+\epsilon_\eta)^{1/3}\Big) + \mathcal{O}\left(\frac{\big(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\big)^{2/3}}{T^{2/3}}\right).
\end{aligned} \tag{61}$$

C.1 Auxiliary Lemmas

LEMMA 2 (SMOOTHNESS OF THE MODIFIED OBJECTIVE). *Under Assumption 2 the modified local objectives $G_1(\mathbf{x})\cdots G_N(\mathbf{x})$ are Lipschitz smooth for any $\mathbf{x}$ with the smoothness parameter given by $\frac{\gamma a_{\max} L_c^2}{6\sqrt{3}} + ((1-\gamma) + \gamma a_{\max} q_i(\mathbf{x}))L_s \le \frac{\gamma a_{\max} L_c^2}{6\sqrt{3}} + ((1-\gamma) + \frac{\gamma a_{\max}}{4})L_s = \frac{\gamma a_{\max} L_c^2}{6\sqrt{3}} + \frac{w_{\max}}{4}L_s = \tilde{L}_s$.*

PROOF. Let $\|.\|_{op}$ denote the spectral norm of the matrix. We need to show a finite upper bound on the operator norm of Hessian of G for $\tilde{L}_s$ smoothness. At any point $\mathbf{x}$, for $G_i(\mathbf{x}) = ((1-\gamma)F_i(\mathbf{x}) + \gamma a_i \sigma_i(\mathbf{x}))$ we have

$$\begin{aligned}
&\|\nabla^2 G_i(\mathbf{x})\|_{op} = \|(1-\gamma)\nabla^2 F_i(\mathbf{x}) + \gamma a_i \nabla^2\big(\sigma_i(\mathbf{x})\big)\|_{op} \\
&= \|\gamma a_i \sigma(\mathbf{x})(1-\sigma(\mathbf{x}))(1-2\sigma(\mathbf{x}))(\nabla F_i(\mathbf{x})\nabla F_i(\mathbf{x})^\top) + ((1-\gamma) + \gamma a_i \sigma(\mathbf{x})(1-\sigma(\mathbf{x})))\nabla^2 F_i(\mathbf{x})\|_{op} \\
&\le \gamma a_i \sigma(\mathbf{x})(1-\sigma(\mathbf{x}))(1-2\sigma(\mathbf{x}))\left\|\nabla F_i(\mathbf{x})\nabla F_i(\mathbf{x})^\top\right\|_{op} + ((1-\gamma) + \gamma a_i \sigma(\mathbf{x})(1-\sigma(\mathbf{x})))\|\nabla^2 F_i(\mathbf{x})\|_{op} \\
&\le \frac{\gamma a_i L_c^2}{6\sqrt{3}} + ((1-\gamma) + \gamma a_i q_i(\mathbf{x}))L_s.
\end{aligned}$$

Replacing a_i by $a_{\max}$ gives us a uniform bound for all $i \in [N]$. Further bounding $\gamma a_{\max}$ by $w_{\max}$ we can bound the L-smoothness parameter for the global objective G by

$$\frac{w_{\max} L_c^2}{6\sqrt{3}} + \frac{L_s}{N}\sum_{i=1}^{N}((1-\gamma) + \gamma a_i q_i(\mathbf{x})). \tag{62}$$

□

PROPOSITION 5 (BOUNDED DISSIMILARITY OF $G(\mathbf{x})$). *Under Assumptions 2 and 3 there exist constants $\tilde{\beta}$ and $\tilde{\kappa}$ such that*

$$\frac{1}{N}\sum_{i=1}^{N}\|\nabla G_i(\mathbf{x})\|^2 \le \tilde{\beta}^2\|\nabla G(\mathbf{x})\|^2 + \tilde{\kappa}^2. \tag{63}$$

PROOF. One can show that

$$\frac{1}{N}\sum_{i=1}^{N}\|\nabla G_i(\mathbf{x})\|^2 = \frac{1}{N}\sum_{i=1}^{N}((1-\gamma)+\gamma a_i q_i(\mathbf{x}))^2\,\|\nabla F_i(\mathbf{x})\|^2 \tag{64}$$

$$\leq \frac{w_{\max}^2}{16N}\sum_{i=1}^{N}\|\nabla F_i(\mathbf{x})\|^2 \tag{65}$$

$$\leq \beta^2\left\|\frac{w_{\max}}{4}\nabla F(\mathbf{x})\right\|^2 + \frac{w_{\max}^2\kappa^2}{16} \tag{66}$$

$$\leq 2\beta^2\|\nabla G(\mathbf{x})\|^2 + 2\beta^2\left\|\nabla G(\mathbf{x}) - \frac{w_{\max}}{4}\nabla F(\mathbf{x})\right\|^2 + \frac{w_{\max}^2\kappa^2}{16} \tag{67}$$

$$\leq 2\beta^2\|\nabla G(\mathbf{x})\|^2 + \frac{\beta^2 w_{\max}^2 L_c^2}{4} + \frac{w_{\max}^2\kappa^2}{16}. \tag{68}$$

Thus the inequality will always hold for $\tilde{\beta} = 2\beta^2$ and $\tilde{\kappa}^2 = \frac{\beta^2 w_{\max}^2 L_c^2}{4} + \frac{w_{\max}^2\kappa^2}{16}$. □

LEMMA 3 (LEARNING RATE ESTIMATION ERROR). *For τ_η rounds of gossip averaging used for estimating $s_i^{(t)}$ we have*

$$\frac{1}{N}\sum_{i=1}^{N}\left\|\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right\|^2 (b_i^{(t)})^2 \leq \min\left\{\frac{(\eta)^2(1-p)^{\tau_\eta}N^2}{4}, \frac{w_{\max}^2(\tilde{\eta}^{(t)})^2(1-p)^{\tau_\eta}N^2}{64}\right\} \tag{69}$$

$$= \min\left\{(\eta)^2\epsilon_\eta, \frac{(\tilde{\eta}^{(t)})^2 w_{\max}^2\epsilon_\eta}{16}\right\} \tag{70}$$

where $\epsilon_\eta = \frac{(1-p)^{\tau_\eta}N^2}{4}$. Note that ϵ_η can be made arbitrarily small as required by increasing τ_η.

PROOF. We have $\tilde{\eta}_i^{(t)} = \frac{\eta}{s_i^{(t)}+\epsilon}$ and $\tilde{\eta}^{(t)} = \frac{\eta}{\bar{s}^{(t)}+\epsilon}$ where $s_i^{(t)} = \max\{s_i^{(t,\tau_\eta)}, s_i^{(t,0)}/N\}$, $s_i^{(t,0)} = Nb_i^{(t)}$, and $s_i^{(t,r+1)} = \sum_{j\in\mathcal{N}_i} w_{ij}\, s_j^{(t,r)}$, or equivalently $\mathbf{s}^{(t,r+1)} = \mathbf{W}\mathbf{s}^{(t,r)}$, with $\mathbf{s}^{(t,r)} = [s_1^{(t,r)}\cdots s_N^{(t,r)}]$. Also, $\bar{s}^{(t)} = \frac{1}{N}\sum_{i=1}^{N} s_i^{(t,r)}$ is the same for all r since gossip averaging preserves the sum, thus $\bar{s}^{(t)} = \frac{1}{N}\sum_{i=1}^{N} s_i^{(t,0)} = \sum_{i=1}^{N} b_i^{(t)}$. We bound the estimation error as follows.

$$\frac{1}{N}\sum_{i=1}^{N}\left\|\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right\|^2 (b_i^{(t)})^2 = \frac{\left\|\tilde{\eta}^{(t)}\right\|^2}{N}\sum_{i=1}^{N}(b_i^{(t)})^2\left\|\frac{\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}}{\tilde{\eta}^{(t)}}\right\|^2 \tag{71}$$

$$= \frac{\left\|\tilde{\eta}^{(t)}\right\|^2}{N}\sum_{i=1}^{N}(b_i^{(t)})^2\frac{\left\|\frac{1}{s_i^{(t)}+\epsilon} - \frac{1}{\bar{s}^{(t)}+\epsilon}\right\|^2}{\left\|\frac{1}{\bar{s}^{(t)}+\epsilon}\right\|^2} \tag{72}$$

$$= \frac{\left\|\tilde{\eta}^{(t)}\right\|^2}{N}\sum_{i=1}^{N}(b_i^{(t)})^2\frac{\left\|s_i^{(t)} - \bar{s}^{(t)}\right\|^2}{\left\|s_i^{(t)} + \epsilon\right\|^2} \tag{73}$$

$$\leq \frac{\left\|\tilde{\eta}^{(t)}\right\|^2}{N}\sum_{i=1}^{N}\left\|s_i^{(t)} - \bar{s}^{(t)}\right\|^2 \leq \frac{\left\|\tilde{\eta}^{(t)}\right\|^2}{N}\sum_{i=1}^{N}\left\|s_i^{(t,\tau_\eta)} - \bar{s}^{(t)}\right\|^2 \tag{74}$$

$$\leq \frac{(1-p)^{\tau_\eta}\left\|\tilde{\eta}^{(t)}\right\|^2}{N}\sum_{i=1}^{N}\left\|s_i^{(t,0)} - \bar{s}^{(t)}\right\|^2 \tag{75}$$

$$\leq \frac{(1-p)^{\tau_\eta}\left\|\tilde{\eta}^{(t)}\right\|^2(\max_i s_i^{(t,0)})^2}{4} \tag{76}$$

$$\leq \min\left\{\frac{(\eta)^2(1-p)^{\tau_\eta}N^2}{4}, \frac{w_{\max}^2(\tilde{\eta}^{(t)})^2(1-p)^{\tau_\eta}N^2}{64}\right\} \tag{77}$$

$$= \min\left\{(\eta)^2\epsilon_\eta, \frac{(\tilde{\eta}^{(t)})^2 w_{\max}^2\epsilon_\eta}{16}\right\} \tag{78}$$

Here equation 74 follows since $s_i^{(t)} \geq b_i^{(t)} = s_i^{(t,0)}/N$ and for the second part note if $s_i^{(t)} = s_i^{(t,\tau_\eta)}$ we have $\|s_i^{(t)} - \bar{s}^{(t)}\|^2 = \|s_i^{(t,\tau_\eta)} - \bar{s}^{(t)}\|^2$ and if $s_i^{(t)} = s_i^{(t,0)}/N$, we have $s_i^{(t,\tau_\eta)} \leq s_i^{(t,0)}/N \leq \bar{s}^{(t)}$. since $\bar{s}^{(t)} \geq b_i^{(t)} = s_i^{(t,0)}/N$, hence $\|s_i^{(t)} - \bar{s}^{(t)}\|^2 \leq \|s_i^{(t,\tau_\eta)} - \bar{s}^{(t)}\|^2$ in both cases. Equation (76) follows from $\sum_{i=1}^N \|c_i - \bar{c}\|^2 \leq \frac{N \max_i (c_i)^2}{4}$ for $\bar{c} = \frac{1}{N}\sum_{i=1}^N c_i,\ c_i \geq 0$ and equation (77) follows from $\tilde{\eta}^{(t)} s_i^{(t,0)} = N\left(\frac{\eta}{s_i^{(t)}+\epsilon}\right) b_i^{(t)} \leq N\eta\ \forall i$. □

LEMMA 4 (FRACTIONAL LEARNING RATE ESTIMATION ERROR). *For τ_η rounds of gossip averaging used for estimating $s_i^{(t)}$ we have*

$$\frac{1}{N}\sum_{i=1}^N \frac{\left\|\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right\|^2}{\|\tilde{\eta}^{(t)}\|^2}(b_i^{(t)})^2 \leq \frac{(1-p)^{\tau_\eta} N^2 \max_i (b_i^{(t)})^2}{4} = \max_i (b_i^{(t)})^2 \epsilon_\eta \tag{79}$$

where $\epsilon_\eta = \frac{(1-p)^{\tau_\eta} N^2}{4}$. Note that ϵ_η can be made arbitrarily small as required by increasing τ_η.

PROOF. Proceeding similarly as lemma 3 until equation (76) we have

$$\frac{1}{N}\sum_{i=1}^N \frac{\left\|\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right\|^2}{\|\tilde{\eta}^{(t)}\|^2}(b_i^{(t)})^2 \leq \frac{(1-p)^{\tau_\eta} (\max_i s_i^{(t,0)})^2}{4} = \frac{(1-p)^{\tau_\eta} N^2 \max_i (b_i^{(t)})^2}{4} = \max_i (b_i^{(t)})^2 \epsilon_\eta. \tag{80}$$

□

LEMMA 5 (BOUNDING DRIFT). *For client i, the accumulated local drift error over τ local steps in Algorithm 2 can be bounded as:*

$$\begin{aligned}\sum_{r=1}^{\tau} \mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)} - \mathbf{x}_i^{(t,r)}\right\|^2\right] &\leq 4(\tilde{\eta}^{(t)})^2 (b_i^{(t)})^2 \tau \mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right] + 2\tau L_c^2 \mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right) b_i^{(t)}\right\|^2\right] \\ &\quad + 6\mathbb{E}\left[(\tilde{\eta}^{(t)})^2 (b_i^{(t)})^2 \sigma^2\right] + 6\mathbb{E}\left[(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})^2 (b_i^{(t)})^2 \sigma^2\right].\end{aligned} \tag{81}$$

PROOF.

$$\tau^2 \mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)} - \mathbf{x}_i^{(t,r)}\right\|^2\right] = \mathbb{E}\left[\left\|\tilde{\eta}_i^{(t)}\, b_i^{(t)} \sum_{l=0}^{r-1} \nabla F_i\left(\mathbf{x}_i^{(t,l)}, \xi_i^{(t,l)}\right)\right\|^2\right] \tag{82}$$

$$\begin{aligned}&\leq 3\mathbb{E}\left[\left\|\tilde{\eta}^{(t)}\, b_i^{(t)} \sum_{l=0}^{r-1} \nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right] + 3\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right) b_i^{(t)} \sum_{l=0}^{r-1} \nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right] \\ &\quad + 3\mathbb{E}\left[\left\|\tilde{\eta}_i^{(t)}\, b_i^{(t)} \sum_{l=0}^{r-1} \left(\nabla F_i\left(\mathbf{x}_i^{(t,l)}, \xi_i^{(t,l)}\right) - \nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right)\right\|^2\right]\end{aligned} \tag{83}$$

$$\begin{aligned}&\leq 3\mathbb{E}\left[\left\|\tilde{\eta}^{(t)}\, b_i^{(t)} \sum_{l=0}^{r-1} \nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right] + 3r^2 L_c^2 \mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right) b_i^{(t)}\right\|^2\right] \\ &\quad + 3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2 (b_i^{(t)})^2 \sum_{l=0}^{r-1} \underbrace{\left\|\left(\nabla F_i\left(\mathbf{x}_i^{(t,l)}, \xi_i^{(t,l)}\right) - \nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right)\right\|^2}_{\leq \sigma^2}\right]\end{aligned} \tag{84}$$

$$\leq \underbrace{3\mathbb{E}\left[\left\|\tilde{\eta}^{(t)}\, b_i^{(t)} \sum_{l=0}^{r-1} \nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right]}_{T_7} + 3r^2 L_c^2 \mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right) b_i^{(t)}\right\|^2\right] + 3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2 (b_i^{(t)})^2 r\sigma^2\right] \tag{85}$$

Here equation (85) follows from proposition (4). We can bound T_7 as

$$T_7 = 3(\tilde{\eta}^{(t)})^2 (b_i^{(t)})^2 \mathbb{E}\left[\left\|\sum_{l=0}^{r-1} \nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right] \tag{86}$$

$$\leq r \sum_{l=0}^{r-1} 3(\tilde{\eta}^{(t)})^2 (b_i^{(t)})^2 \mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right] \tag{87}$$

$$\leq r\sum_{l=0}^{r-1} 6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\left(\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]+\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)-\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]\right) \tag{88}$$

$$\leq r\sum_{l=0}^{r-1} 6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\left(\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]+L_s^2\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,l)}-\mathbf{x}_i^{(t,0)}\right\|^2\right]\right) \tag{89}$$

$$\leq 6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\left(r^2\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]+r\sum_{l=0}^{r-1}L_s^2\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,l)}-\mathbf{x}_i^{(t,0)}\right\|^2\right]\right) \tag{90}$$

$$\leq 6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\left(r^2\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]+\tau\sum_{l=0}^{\tau-1}L_s^2\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,l)}\right\|^2\right]\right). \tag{91}$$

Substituting equation (91) into equation (85) we have

$$\begin{aligned}\tau^2\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] &\leq 6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\left(r^2\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]+\tau\sum_{l=0}^{\tau-1}L_s^2\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,l)}\right\|^2\right]\right)\\ &\quad+3r^2L_c^2\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]+3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2r\sigma^2\right].\end{aligned} \tag{92}$$

Summing over τ

$$\begin{aligned}\tau^2\left(1-6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2L_s^2\right)\sum_{i=1}^{\tau}\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] &\leq \sum_{r=1}^{\tau}6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2r^2\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]\\ &\quad+\sum_{r=1}^{\tau}3r^2L_c^2\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]+\sum_{r=1}^{\tau}3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2r\sigma^2\right]\end{aligned} \tag{93}$$

$$\leq 2(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\tau^3\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]+\tau^3L_c^2\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]+\frac{3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2\tau^2\sigma^2\right]}{2}. \tag{94}$$

Since $\tilde{\eta}^{(t)}(b_i^{(t)})^2 \leq \eta$ for $\eta \leq \frac{1}{2\sqrt{3}L_s}$ we have $\frac{1}{(1-6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2L_s^2)} \leq \frac{1}{1-6(\eta)^2L_s^2} \leq 2$ and

$$\begin{aligned}&\sum_{r=1}^{\tau}\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] \leq 4(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\tau\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]\\ &+2\tau L_c^2\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]+3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2\sigma^2\right]\end{aligned} \tag{95}$$

$$\begin{aligned}&\leq 4(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\tau\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]+2\tau L_c^2\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]\\ &+6\mathbb{E}\left[(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\sigma^2\right]+6\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2\sigma^2\right].\end{aligned} \tag{96}$$

□

LEMMA 6 (CONSENSUS RECURSION). *Define* $\Xi_t^2 := \frac{1}{N}\sum_{i=1}^{N}\|\bar{\mathbf{x}}^{(t,0)}-\mathbf{x}_i^{(t,0)}\|^2$. *For* $\tilde{\eta}^{(t)} \leq \frac{\sqrt{p}}{8\sqrt{2}\tilde{L}_s}$ *we have*

$$\Xi_{t+1}^2 \leq \left(1-\frac{p}{4}\right)\Xi_t^2+\frac{32}{p}(\tilde{\eta}^{(t)})^2\tilde{\beta}^2\|\nabla G_i(\overline{\mathbf{x}}^{(t,0)})\|^2+\frac{32}{p}(\tilde{\eta}^{(t)})^2\tilde{\kappa}^2+\frac{(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{2p}\epsilon_\eta+\frac{3(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{4p\tau}(1+\epsilon_\eta). \tag{97}$$

PROOF. Let $\mathbf{H}^{(t)} := \frac{1}{\tau}\left[\mathbf{h}_1^t\cdots\mathbf{h}_N^t\right] \in \mathbb{R}^{d\times N}$ and $\tilde{\boldsymbol{\eta}}^{(t)} := [\tilde{\eta}_1^{(t)}\cdots\tilde{\eta}_N^{(t)}] \in \mathbb{R}^{1\times N}$. We have

$$N\Xi_{t+1}^2 = \mathbb{E}\left\|\tilde{\mathbf{X}}^{(t+1,0)}-\mathbf{X}^{(t+1,0)}\right\|_F^2 = \mathbb{E}\left\|\left(\mathbf{X}^{(t+1,0)}\right)\left(\mathbf{I}-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\right\|_F^2 \tag{98}$$

$$= \mathbb{E}\left\|\left(\mathbf{X}^{(t,0)}-\mathbf{H}(\tilde{\boldsymbol{\eta}}^{(t)})^T\right)W\left(\mathbf{I}-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\right\|_F^2 \tag{99}$$

$$= \mathbb{E}\left\|\left(\mathbf{X}^{(t,0)}-\mathbf{H}(\tilde{\boldsymbol{\eta}}^{(t)})^T\right)\left(\mathbf{I}-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\left(W-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\right\|_F^2 \tag{100}$$

$$\leq (1-p)\mathbb{E}\left\|\left(\mathbf{X}^{(t,0)}-\mathbf{H}(\tilde{\boldsymbol{\eta}}^{(t)})^T\right)\left(\mathbf{I}-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\right\|_F^2 \tag{101}$$

$$\leq (1-p)\left(1+\frac{p}{2}\right)\mathbb{E}\left\|\mathbf{X}^{(t,0)}\left(\mathbf{I}-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\right\|_F^2 + (1-p)\left(1+\frac{2}{p}\right)\mathbb{E}\left\|\left(\mathbf{H}(\tilde{\boldsymbol{\eta}}^{(t)})^T\right)\left(\mathbf{I}-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\right\|_F^2 \tag{102}$$

$$\leq \left(1-\frac{p}{2}\right)N\Xi_t^2 + \frac{2}{p}\mathbb{E}\left\|\left(\mathbf{H}(\tilde{\boldsymbol{\eta}}^{(t)})^T\right)\right\|_F^2 \tag{103}$$

$$= \left(1-\frac{p}{2}\right)N\Xi_t^2 + \frac{2}{p}\underbrace{\frac{1}{\tau^2}\sum_{i=1}^{N}\mathbb{E}\left\|\tilde{\eta}_i^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)},\xi_i^{(t,r)})\right\|^2}_{T_8}. \tag{104}$$

Here equation (103) follows from $\|AB\|_F \leq \|A\|_F\|B\|_2$ and $\left\|\left(\mathbf{I}-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\right\|_2 \leq 1$. We bound T_8 as

$$T_8 = \frac{1}{\tau^2}\sum_{i=1}^{N}\mathbb{E}\left\|\tilde{\eta}_i^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)},\xi_i^{(t,r)})\right\|^2 \tag{105}$$

$$\leq 3\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|\tilde{\eta}^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)})\right\|^2 + 3\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)})b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)})\right\|^2 \tag{106}$$

$$+ 3\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|\tilde{\eta}_i^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\left(\nabla F_i(\mathbf{x}_i^{(t,r)},\xi_i^{(t,r)})-\nabla F_i(\mathbf{x}_i^{(t,r)})\right)\right\|^2$$

$$\leq 3\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|\tilde{\eta}^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)})\right\|^2 + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta + \frac{3}{\tau}\sum_{i=1}^{N}\left\|\tilde{\eta}_i^{(t)}b_i^{(t)}\right\|^2\sigma^2 \tag{107}$$

$$\leq 3\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|\tilde{\eta}^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)})\right\|^2 + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta + \frac{6}{\tau}\sum_{i=1}^{N}\left\|\tilde{\eta}_i^{(t)}b_i^{(t)}\right\|^2\sigma^2 + \frac{6}{\tau}\sum_{i=1}^{N}\left\|(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)})b_i^{(t)}\right\|^2\sigma^2 \tag{108}$$

$$\leq 6\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|\tilde{\eta}^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,0)})\right\|^2 + 6\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|\tilde{\eta}^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\left(\nabla F_i(\mathbf{x}_i^{(t,r)})-\nabla F_i(\mathbf{x}_i^{(t,0)})\right)\right\|^2$$

$$\frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta) \tag{109}$$

$$\leq \sum_{i=1}^{N}6(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + 6\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|\tilde{\eta}^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\left(\nabla F_i(\mathbf{x}_i^{(t,r)})-\nabla F_i(\mathbf{x}_i^{(t,0)})\right)\right\|^2$$

$$+ \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta) \tag{110}$$

$$\leq \sum_{i=1}^{N}6(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{6}{\tau}\sum_{i=1}^{N}\sum_{r=0}^{\tau-1}(\eta)^2\mathbb{E}\left\|\left(\nabla F_i(\mathbf{x}_i^{(t,r)})-\nabla F_i(\mathbf{x}_i^{(t,0)})\right)\right\|^2$$

$$+ \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta) \tag{111}$$

$$\leq \sum_{i=1}^{N}6(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{6}{\tau}\sum_{i=1}^{N}L_s^2(\eta)^2\sum_{r=0}^{\tau-1}\mathbb{E}\left\|\left(\mathbf{x}_i^{(t,r)}-\mathbf{x}_i^{(t,0)}\right)\right\|^2 + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta$$

$$+ \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta) \tag{112}$$

$$\leq \sum_{i=1}^{N}6(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta)$$

$$+ \frac{6}{\tau}\sum_{i=1}^{N}L_s^2(\eta)^2\Bigg((4(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\tau\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right] + 2\tau L_c^2\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]$$

$$+ 6\mathbb{E}\left[(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\sigma^2\right] + 6\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2\sigma^2\right]\Bigg) \tag{113}$$

$$\leq \sum_{i=1}^{N}6(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta)$$

$$+ \sum_{i=1}^{N} 24L_s^2(\eta)^2(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\|\nabla G(\mathbf{x}_i^{(t,0)})\|^2\right] + \frac{3NL_s^2(\eta)^2(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{4}\epsilon_\eta + \frac{9N(\eta)^2(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_s^2\sigma^2}{4\tau}(1+\epsilon_\eta). \tag{114}$$

Here equation (107) follows from lemma 3 and proposition (4). Equation (109) also uses lemma 3. Equation (110) follows from the definition of G, equation (111) follows from the fact that $\tilde{\eta}^{(t)} b_i^{(t)} \leq \eta$ and equation (113) follows from lemma 5. Equation (114) also follows from from lemma 3. For $\eta \leq \frac{1}{2\sqrt{3}L_s}$ we can upper bound equation (114) as

$$T_8 \leq \sum_{i=1}^{N} 6(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{16}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{16\tau}(1+\epsilon_\eta)$$

$$+ \sum_{i=1}^{N} 2(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{N(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{16}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{16\tau}(1+\epsilon_\eta) \tag{115}$$

$$= \sum_{i=1}^{N} 8(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{N(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{4}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta) \tag{116}$$

$$\leq \sum_{i=1}^{N} 16(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{N(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{4}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta)$$

$$+ \sum_{i=1}^{N} 16(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)}) - \nabla G_i(\overline{\mathbf{x}}^{(t,0)})\|^2 \tag{117}$$

$$\leq \sum_{i=1}^{N} 16(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{N(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{4}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta)$$

$$+ \sum_{i=1}^{N} 16(\tilde{\eta}^{(t)})^2\tilde{L}_s^2\|\mathbf{x}_i^{(t,0)} - \overline{\mathbf{x}}^{(t,0)})\|^2 \tag{118}$$

$$= 16\sum_{i=1}^{N}(\tilde{\eta}^{(t)})^2\|\nabla G_i(\overline{\mathbf{x}}^{(t,0)})\|^2 + \frac{N(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{4}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta) + 16N(\tilde{\eta}^{(t)})^2\tilde{L}_s^2\Xi_t^2 \tag{119}$$

$$\leq 16N\tilde{\beta}^2(\tilde{\eta}^{(t)})^2\|\nabla G(\overline{\mathbf{x}}^{(t,0)})\|^2 + 16N(\tilde{\eta}^{(t)})^2\tilde{\kappa}^2 + \frac{N(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{4}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta)$$

$$+ 16N(\tilde{\eta}^{(t)})^2\tilde{L}_s^2\Xi_t^2. \tag{120}$$

Substituting back in equation (104) we have

$$\Xi_{t+1}^2 \leq \left(1 - \frac{p}{2}\right)\Xi_t^2 + \frac{2}{p}\Bigg(16(\tilde{\eta}^{(t)})^2\tilde{\beta}^2\|\nabla G(\overline{\mathbf{x}}^{(t,0)})\|^2 + 16(\tilde{\eta}^{(t)})^2\tilde{\kappa}^2 \tag{121}$$

$$+ \frac{(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{4}\epsilon_\eta + \frac{3(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta) + 16(\tilde{\eta}^{(t)})^2\tilde{L}_s^2\Xi_t^2\Bigg). \tag{122}$$

For $\tilde{\eta}^{(t)} \leq \frac{\sqrt{p}}{8\sqrt{2}\tilde{L}_s}$ we have

$$\Xi_{t+1}^2 \leq \left(1 - \frac{p}{4}\right)\Xi_t^2 + \frac{32}{p}(\tilde{\eta}^{(t)})^2\tilde{\beta}^2\|\nabla G_i(\overline{\mathbf{x}}^{(t,0)})\|^2 + \frac{32}{p}(\tilde{\eta}^{(t)})^2\tilde{\kappa}^2 + \frac{(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{2p}\epsilon_\eta + \frac{3(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{4p\tau}(1+\epsilon_\eta). \tag{123}$$

□

LEMMA 7 (CONSENSUS CONTROL). *For $\tilde{\eta}^{(t)} \leq \frac{\sqrt{p}}{8\sqrt{2}\tilde{L}_s}$ the term $\frac{1}{T}\sum_{t=0}^{T-1}\Xi_t^2$ can be bounded as*

$$\frac{1}{T}\sum_{t=0}^{T-1}\Xi_t^2 \leq \frac{4\Xi_0^2}{pT} + \frac{128}{p^2T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,\tilde{\beta}^2\,\|\nabla G_i(\overline{\mathbf{x}}^{(t,0)})\|^2$$

$$+ \frac{128}{p^2T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,\tilde{\kappa}^2 + \frac{2w_{\max}^2}{p^2T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,L_c^2\,\epsilon_n + \frac{3w_{\max}^2}{p^2\tau T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,\sigma^2(1+\epsilon_\eta). \tag{124}$$

PROOF. Consider the recursion equation in lemma 6. We have

$$\Xi_{t+1}^2 \leq \left(1 - \frac{p}{4}\right)\Xi_t^2 + A_t \tag{125}$$

where

$$A_t = \left(1 - \frac{p}{4}\right)\Xi_t^2 + \frac{32}{p}(\tilde{\eta}^{(t)})^2\tilde{\beta}^2\|\nabla G_i(\overline{\mathbf{x}}^{(t,0)})\|^2 + \frac{32}{p}(\tilde{\eta}^{(t)})^2\tilde{\kappa}^2 + \frac{(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{2p}\epsilon_\eta + \frac{3(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{4p\tau}(1+\epsilon_\eta). \tag{126}$$

Rearranging and summing over T we have

$$\Xi_T^2 - \Xi_0^2 = \sum_{t=0}^{T-1}\left(\Xi_{t+1}^2 - \Xi_t^2\right) \le -\frac{p}{4}\sum_{t=0}^{T-1}\Xi_t^2 + \sum_{t=0}^{T-1}A_t. \tag{127}$$

Since $\Xi_T^2 \ge 0$

$$\frac{1}{T}\sum_{t=0}^{T-1}\Xi_t^2 \le \frac{4}{pT}\left(\Xi_0^2 + \sum_{t=0}^{T-1}A_t\right). \tag{128}$$

Substituting back A_t gives

$$\begin{aligned}\frac{1}{T}\sum_{t=0}^{T-1}\Xi_t^2 \le{}& \frac{4\Xi_0^2}{pT} + \frac{128}{p^2T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,\tilde{\beta}^2\,\|\nabla G_i(\overline{\mathbf{x}}^{(t,0)})\|^2 \\ &+ \frac{128}{p^2T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,\tilde{\kappa}^2 + \frac{2w_{\max}^2}{p^2T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,L_c^2\,\epsilon_n + \frac{3w_{\max}^2}{p^2\tau T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,\sigma^2(1+\epsilon_\eta).\end{aligned} \tag{129}$$

□

D Modified Objective with Group-wise Penalty

In this section, we define the following terms, which will be used exclusively within this context.

$$\mathbf{X} := [\mathbf{x}_1, \ldots, \mathbf{x}_N] \in \mathbb{R}^{d\times N}$$
$$\bar{\mathbf{X}} := [\bar{\mathbf{x}}, \ldots, \bar{\mathbf{x}}] \in \mathbb{R}^{d\times N},$$

where $\bar{\mathbf{x}} = \frac{1}{N}\sum_{i=1}^N \mathbf{x}_i$. Similarly at the beginning of round t one can define

$$\mathbf{X}^{(t,0)} := \left[\mathbf{x}_1^{(t,0)}, \ldots, \mathbf{x}_N^{(t,0)}\right] \in \mathbb{R}^{d\times N},$$
$$\bar{\mathbf{X}}^{(\mathbf{t},\mathbf{0})} := \left[\bar{\mathbf{x}}^{(t,0)}, \ldots, \bar{\mathbf{x}}^{(t,0)}\right] \in \mathbb{R}^{d\times N},$$

where $\bar{\mathbf{x}}^{(t,0)} = \frac{1}{N}\sum_{i=1}^N \mathbf{x}_i^{(t,0)}$. The modified objective function with group wise penalty is defined as follows

$$G(\mathbf{x}) = \frac{1}{N}\left\{(1-\gamma)\left(\sum_{i=1}^N F_i(\mathbf{x})\right) + \gamma\left(\sum_{k=1}^K \lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)\right)\right\} \tag{130}$$

$$= G(\mathbf{x}) = \frac{1}{N}\left\{(1-\gamma)\left(\sum_{i=1}^N F_i(\mathbf{x})\right) + \gamma\left(\sum_{k=1}^K \lambda_k M_k(\mathbf{x})\right)\right\}. \tag{131}$$

Here $M_k(\mathbf{x}) := \prod_{j\in C_k}\sigma_j(\mathbf{x})$ which tells us about the probability of all nodes in C_k dropping out given a local model $\mathbf{x}$ at each node. . The gradient of the function $\sigma(\mathbf{x})$ is given by $\sigma_i(\mathbf{x})(1-\sigma_i(\mathbf{x}))\nabla F_i(\mathbf{x})$. Note that the gradient of the global objective here is given by

$$\nabla G(\mathbf{x}) = \frac{1}{N}\left(\sum_{i=1}^N(1-\gamma)\nabla F_i(\mathbf{x}) + \gamma\sum_{k=1}^K\sum_{l\in C_k}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_l(\mathbf{x}))\nabla F_l(\mathbf{x})\right). \tag{132}$$

One can rewrite the gradient of the modified objective in equation (9) as

$$\nabla G(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N\left((1-\gamma) + \gamma\sum_{k\in\mathcal{K}_i}\lambda_k M_k(\mathbf{x})(1-\sigma_i(\mathbf{x}))\right)\nabla F_i(\mathbf{x}). \tag{133}$$

Where $\mathcal{K}_i := \{k | i \in C_k\} \subseteq [K]$ is the set of all groups which node i is a part of. For the ease of purpose we define the function $\Psi(\mathbf{X})$ as follows

$$\Psi(\mathbf{X}) := \frac{1}{N}\left((1-\gamma)\left(\sum_{i=1}^N F_i(\mathbf{x})\right) + \gamma\sum_{k=1}^K\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x}_j)\right)\right) \tag{134}$$

and its gradient is given by

$$[\nabla\Psi(\mathbf{X})]_i = \frac{1}{N}\left((1-\gamma) + \gamma\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x}_j)\right)(1-\sigma_i(\mathbf{x}_i))\right)\nabla F_i(\mathbf{x}_i) \tag{135}$$

$$= \frac{1}{N}\left((1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k M_k(\mathbf{X})(1-\sigma_i(\mathbf{x}_i))\right)\nabla F_i(\mathbf{x}_i) \tag{136}$$

where

$$M_k(\mathbf{X}) = \prod_{j\in C_k} \sigma_j(\mathbf{x}_j)$$

. Let $M_{ki}(\mathbf{X})$ be the estimate of $M_k(\mathbf{X})$ at node i given by

$$M_{ki}(\mathbf{X}) = \min\left\{M_{ki}^{(\tau_b)}(\mathbf{X}), \left(M_{ki}^{(0)}(\mathbf{X})\right)^{1/N}\right\} \tag{137}$$

with

$$M_{ki}^{(0)} = \begin{cases} (\sigma_i(\mathbf{x}_i))^N & \text{if } k \in \mathcal{K}_i \\ 1 & \text{if } k \notin \mathcal{K}_i \end{cases} \tag{138}$$

Here $M_{ki}^{(\tau_b)}(\mathbf{X})$ is the value after τ_b rounds of gossip averaging. Additionally $G(\bar{\mathbf{x}}) = \Psi(\bar{\mathbf{X}})$ and $\nabla G(\bar{\mathbf{x}}) = \sum_{i=1}^N \left[\nabla\Psi(\bar{\mathbf{X}})\right]_i$. We also have $M_{ki}^{(t)} = M_{ki}(\mathbf{X}^{(t,0)})$ and $M_k^{(t)} = M_k(\mathbf{X}^{(t,0)})$. Also recall the terms $b_i^{(t)}$ and $\hat{b}_i^{(t)}$

$$\hat{b}_i^{(t)} = (1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k M_k^{(t)}(1-\sigma(\mathbf{x}_i^{(t,0)})) = (1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k M_k(\mathbf{X}^{(t,0)})(1-\sigma(\mathbf{x}_i^{(t,0)})) \tag{139}$$

$$\hat{b}_i^{(t)} = (1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k M_{ki}^{(t)}(1-\sigma(\mathbf{x}_i^{(t,0)})) = (1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k M_{ki}(\mathbf{X}^{(t,0)})(1-\sigma(\mathbf{x}_i^{(t,0)})) \tag{140}$$

where $b_i^{(t)}$ denotes the true pre-factor at node i, and $\hat{b}_i^{(t)}$ represents its estimate at node i obtained after τ_b rounds of log gossip. Next we define the aggregated update, its noiseless/expected value and its expected value under perfect pre-factor estimation as

$$\mathbf{h}_i^t = \hat{b}_i^{(t)} \sum_{r=0}^{\tau-1} \nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)}) \tag{141}$$

$$\widehat{\mathbf{h}}_i^t = \hat{b}_i^{(t)} \sum_{r=0}^{\tau-1} \nabla F_i(\mathbf{x}_i^{(t,r)}) \tag{142}$$

$$\overline{\mathbf{h}}_i^t = b_i^{(t)} \sum_{r=0}^{\tau-1} \nabla F_i(\mathbf{x}_i^{(t,r)}). \tag{143}$$

We also define the constant

$$w_{\max} = 4(1-\gamma) + \gamma \max_{i\in[N]}\Big(\sum_{k\in\mathcal{K}_i} \lambda_k\Big). \tag{144}$$

We will repeatedly use the fact $\hat{b}_i^{(t)} \le \frac{w_{\max}}{4}$ and $b_i^{(t)} \le \frac{w_{\max}}{4}$ (see lemma 10). Define $\tilde{\eta}^{(t)} := \frac{\eta}{\bar{s}^{(t)}+\epsilon}$ and $\tilde{\eta}_i^{(t)} := \frac{\eta}{s_i^{(t)}+\epsilon}$ is the estimate of $\tilde{\eta}^{(t)}$ at node $i \in [N]$. Note that the update at node i during round r can be rewritten as

$$\mathbf{x}_i^{(t,r+1)} := \mathbf{x}_i^{(t,r)} - \hat{\eta}_i^{(t)} \nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)}) = \mathbf{x}_i^{(t,r)} - \tilde{\eta}_i^{(t)} \hat{b}_i^{(t)} \nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})/\tau. \tag{145}$$

From the $\tilde{L}_s$ smoothness of the global objective function (see lemma 8)

$$\mathbb{E}\left[G(\bar{\mathbf{x}}^{(t+1,0)})\right] - \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t,0)})\right] \le \underbrace{-\mathbb{E}\left[\left\langle \nabla G(\bar{\mathbf{x}}^{(t,0)}), \left(\frac{1}{\tau N}\sum_{i=1}^N \tilde{\eta}_i^{(t)} \mathbf{h}_i^t\right)\right\rangle\right]}_{T_9} + \underbrace{\mathbb{E}\left[\frac{\tilde{L}_s}{2}\left\|\left(\frac{1}{\tau N}\sum_{i=1}^N \tilde{\eta}_i^{(t)} \mathbf{h}_i^t\right)\right\|^2\right]}_{T_{10}}. \tag{146}$$

We first focus on simplifying T_{10}

$$T_{10} = \mathbb{E}\left[\frac{\tilde{L}_s}{2}\left\|\left(\frac{1}{\tau N}\sum_{i=1}^N \tilde{\eta}_i^{(t)} \mathbf{h}_i^t\right)\right\|^2\right] \tag{147}$$

$$= \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2}{2}\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^N \frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}} \mathbf{h}_i^t\right)\right\|^2\right] \tag{148}$$

$$\leq \tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\widehat{\mathbf{h}}_i^t\right)\right\|^2\right] + \underbrace{\tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\left(\mathbf{h}_i^t-\widehat{\mathbf{h}}_i^t\right)\right)\right\|^2\right]}_{T_{11}}. \tag{149}$$

T_{11} here can be bounded as

$$T_{11} = \tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\left(\mathbf{h}_i^t-\widehat{\mathbf{h}}_i^t\right)\right)\right\|^2\right] \tag{150}$$

$$= \frac{\tilde{L}_s}{\tau^2N^2}\mathbb{E}\left[\sum_{i=1}^{N}(\tilde{\eta}_i^{(t)})^2\left\|\left(\hat{b}_i^{(t)}\sum_{r=0}^{\tau-1}\left(\nabla F_i(\mathbf{x}_i^{(t,r)},\xi_i^{(t,r)})-F_i(\mathbf{x}_i^{(t,r)})\right)\right)\right\|^2\right] \tag{151}$$

$$\leq \frac{\tilde{L}_s}{\tau N^2}\sum_{i=1}^{N}(\tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2 \tag{152}$$

$$\leq \frac{2\tilde{L}_s}{\tau N^2}\sum_{i=1}^{N}(\tilde{\eta}^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2 + \frac{2\tilde{L}_s}{\tau N^2}\sum_{i=1}^{N}\|\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)}\|^2(\hat{b}_i^{(t)})^2\sigma^2 \tag{153}$$

$$\leq \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau N} + \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2w_{\max}\sigma^2\epsilon_\eta}{8\tau N}. \tag{154}$$

Here equation 154 follows from lemma 11 and lemma 10. Equation (152) follows from proposition 4 and equation 151 follows from independent sampling of mini-batches by the clients. Thus we have

$$T_{10} \leq \tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\widehat{\mathbf{h}}_i^t\right)\right\|^2\right] + \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau N}(1+\epsilon_\eta). \tag{155}$$

Consider the term T_9

$$T_9 = -\mathbb{E}\left[\left\langle\nabla G(\bar{\mathbf{x}}^{(t,0)}),\left(\frac{1}{\tau N}\sum_{i=1}^{N}\tilde{\eta}_i^{(t)}\widehat{\mathbf{h}}_i^t\right)\right\rangle\right] - \mathbb{E}\left[\left\langle\nabla G(\bar{\mathbf{x}}^{(t,0)}),\left(\frac{1}{\tau N}\sum_{i=1}^{N}\tilde{\eta}_i^{(t)}(\mathbf{h}_i^t-\widehat{\mathbf{h}}_i^t)\right)\right\rangle\right] \tag{156}$$

$$= -\frac{\tilde{\eta}^{(t)}}{2}\left\|\nabla G(\bar{\mathbf{x}}^{(t,0)})\right\|^2 - \underbrace{\frac{\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\widehat{\mathbf{h}}_i^t\right\|^2}_{T_{12}} + \underbrace{\frac{\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\nabla G(\bar{\mathbf{x}}^{(t,0)}) - \frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\widehat{\mathbf{h}}_i^t\right\|^2}_{T_{13}}. \tag{157}$$

T_{13} her an be further bounded as

$$T_{13} = \frac{\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\nabla G(\bar{\mathbf{x}}^{(t,0)}) - \frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\widehat{\mathbf{h}}_i^t\right\|^2 \tag{158}$$

$$= \frac{\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\sum_{i=1}^{N}\left[\nabla\Psi(\bar{\mathbf{X}}^{(t,0)})\right]_i - \frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\widehat{\mathbf{h}}_i^t\right\|^2 \tag{159}$$

$$\begin{aligned}
&\leq 2\tilde{\eta}^{(t)}\mathbb{E}\left[\left\|\sum_{i=1}^{N}\left(\left[\nabla\Psi(\bar{\mathbf{X}}^{(t,0)})\right]_i - \left[\nabla\Psi(\mathbf{X}^{(t,0)})\right]_i\right)\right\|^2\right] + 2\tilde{\eta}^{(t)}\mathbb{E}\left[\left\|\sum_{i=1}^{N}\left[\Psi(\mathbf{X}^{(t,0)})\right]_i - \frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\right\|^2\right] \\
&\quad + 2\tilde{\eta}^{(t)}\mathbb{E}\left[\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t - \frac{1}{\tau N}\sum_{i=1}^{N}\widehat{\mathbf{h}}_i^t\right\|^2\right] + 2\tilde{\eta}^{(t)}\left[\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\widehat{\mathbf{h}}_i^t\left(1-\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\right)\right\|^2\right]
\end{aligned} \tag{160}$$

$$\leq \underbrace{2\tilde{\eta}^{(t)}\tilde{L}_g^2\left\|\bar{\mathbf{X}}^{(t,0)}-\mathbf{X}^{(t,0)}\right\|_F^2}_{\text{Consensus error}} + \underbrace{2\tilde{\eta}^{(t)}\mathbb{E}\left[\left\|\sum_{i=1}^{N}\left[\Psi(\mathbf{X}^{(t,0)})\right]_i - \frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\right\|^2\right]}_{T_{14}}$$

$$+ \underbrace{2\tilde{\eta}^{(t)}\mathbb{E}\left[\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t - \frac{1}{\tau N}\sum_{i=1}^{N}\widehat{\mathbf{h}}_i^t\right\|^2\right]}_{T_{15}} + \underbrace{2\tilde{\eta}^{(t)}\left[\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\widehat{\mathbf{h}}_i^t\left(1-\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\right)\right\|^2\right]}_{T_{16}}. \tag{161}$$

Here equation (161) follows from Jensen's inequality and lemma 9. T_{15} and T_{16} can be bounded using lemma 14 and lemma 12 respectively

$$T_{15} = \frac{2\tilde{\eta}^{(t)}}{N^2\tau^2}\mathbb{E}\left[\left\|\sum_{i=1}^{N}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)})\left(b_i^{(t)} - \hat{b}_i^{(t)}\right)\right\|^2\right] \tag{162}$$

$$\leq \frac{2\tilde{\eta}^{(t)}L_c^2}{N\tau}\mathbb{E}\left[\sum_{r=0}^{\tau-1}\sum_{i=1}^{N}\left\|b_i^{(t)} - \hat{b}_i^{(t)}\right\|^2\right] \tag{163}$$

$$\leq 2\tilde{\eta}^{(t)}L_c^2\epsilon_b \tag{164}$$

where $K_{\max} = \max_i |\mathcal{K}_i|$. Here equation (164) follows from lemma 14. Now T_{16} can be bounded as

$$T_{16} = 2\tilde{\eta}^{(t)}\mathbb{E}\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\hat{b}_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)})\left(1-\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\right)\right\|^2 \tag{165}$$

$$\leq \frac{2\tilde{\eta}^{(t)}}{\tau N}\sum_{i=1}^{N}\sum_{r=0}^{\tau-1}\left\|\hat{b}_i^{(t)}\nabla F_i(\mathbf{x}_i^{(t,r)})\left(1-\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\right)\right\|^2 \tag{166}$$

$$\leq \frac{2\tilde{\eta}^{(t)}L_c^2}{N}\sum_{i=1}^{N}\frac{\|\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\|^2}{\|\tilde{\eta}^{(t)}\|^2}(\hat{b}_i^{(t)})^2 \tag{167}$$

$$\leq \frac{\tilde{\eta}^{(t)}L_c^2 w_{\max}^2}{8}\epsilon_\eta. \tag{168}$$

Equation (168) follows from lemma 12. Next we note that using the definition in equation (140) the term $\left[\Psi(\mathbf{X}^{(t,0)})\right]_i$ can be rewritten as

$$\left[\nabla\Psi(\mathbf{X}^{(t,0)})\right]_i = \frac{1}{N}\left((1-\gamma) + \gamma\sum_{k\in\mathcal{K}_i}\lambda_k M_k(\mathbf{X}^{(\mathbf{t},\mathbf{0})})(1-\sigma_i(\mathbf{x}_i^{(t,0)}))\right)\nabla F_i(\mathbf{x}_i^{(t,0)})$$

$$= \frac{b_i^{(t)}}{N}\nabla F_i(\mathbf{x}_i^{(t,0)}).$$

Plugging this into T_{14} and using the definitions in equations (136),(143) we have

$$T_{14} = 2\tilde{\eta}^{(t)}\mathbb{E}\left[\left\|\sum_{i=1}^{N}\left[\Psi(\mathbf{X}^{(t,0)})\right]_i - \frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\right\|^2\right] \tag{169}$$

$$= 2\tilde{\eta}^{(t)}\mathbb{E}\left[\left\|\frac{1}{\tau N}\sum_{i=1}^{N}b_i^{(t)}\sum_{r=0}^{\tau-1}\left(\nabla F_i(\mathbf{x}_i^{(t,0)}) - \nabla F_i(\mathbf{x}_i^{(t,r)})\right)\right\|^2\right] \tag{170}$$

$$\leq \frac{2\tilde{\eta}^{(t)}}{N\tau}\sum_{i=1}^{N}(b_i^{(t)})^2\sum_{r=0}^{\tau-1}\mathbb{E}\left[\left\|\nabla F_i(\mathbf{x}_i^{(t,0)}) - \nabla F_i(\mathbf{x}_i^{(t,r)})\right\|^2\right] \tag{171}$$

$$\leq \frac{2\tilde{\eta}^{(t)}L_s^2}{N\tau}\sum_{i=1}^{N}(b_i^{(t)})^2\sum_{r=0}^{\tau-1}\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)} - \mathbf{x}_i^{(t,r)}\right\|^2\right]. \tag{172}$$

Substituting the upper bound from lemma (15) into equation (172) we can further bound T_{14} as

$$T_{14} \leq \frac{2\tilde{\eta}^{(t)}L_s^2}{N\tau}\sum_{i=1}^{N}(\hat{b}_i^{(t)})^2\left(\tau L_c^2\left((\tilde{\eta}^{(t)})^2(b_i^{(t)})^2 + \mathbb{E}\left[\left\|(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\right\|^2\right]\right)\right.$$
$$\left. + \frac{3(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{16} + 3\mathbb{E}\left[(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2\right]\right) \tag{173}$$

$$\leq \frac{(\tilde{\eta}^{(t)})^3 w_{\max}^4 L_s^2 L_c^2}{128} + \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^4 L_s^2\sigma^2}{128\tau} + \frac{3\tilde{\eta}^{(t)}w_{\max}^2 L_s^2\sigma^2}{8N\tau}\sum_{i=1}^{N}E\left[(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\right]$$

$$+ \frac{\tilde{\eta}^{(t)} w_{\max}^2 L_s^2 L_c^2}{8N} \sum_{i=1}^{N} \mathbb{E}\left[\left\|(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})\, \hat{b}_i^{(t)}\right\|^2\right] \tag{174}$$

$$\leq \frac{(\tilde{\eta}^{(t)})^3 w_{\max}^4 L_s^2 L_c^2}{128}(1+\epsilon_\eta) + \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^4 L_s^2 \sigma^2}{128\tau}(1+\epsilon_\eta). \tag{175}$$

Here equation (174) follows from lemma 10 and equation (175) follows from lemma11. Now using equations (146), (155), (157), (161), (164), (168) and (175) we have

$$\mathbb{E}\left[G(\bar{\mathbf{x}}^{(t+1,0)})\right] - \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t,0)})\right] \leq -\frac{\tilde{\eta}^{(t)}}{2}\left\|\nabla G(\overline{\mathbf{x}}^{(t,0)})\right\|^2 + \tilde{\eta}^{(t)}\left(\tilde{L}_s\tilde{\eta}^{(t)} - \frac{1}{2}\right)\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\widehat{\mathbf{h}}_i^t\right)\right\|^2\right]$$
$$+ 2\tilde{\eta}^{(t)} N \tilde{L}_g^2 \Xi_t^2 + \tilde{\eta}^{(t)}\left(2L_c^2\epsilon_b + \frac{L_c^2 w_{\max}^2}{8}\epsilon_\eta\right) + (\tilde{\eta}^{(t)})^2\left(\frac{\tilde{L}_s w_{\max}^2 \sigma^2}{8\tau N}(1+\epsilon_\eta)\right)$$
$$+ (\tilde{\eta}^{(t)})^3\left(\frac{w_{\max}^4 L_s^2 L_c^2}{128}(1+\epsilon_\eta) + \frac{3w_{\max}^4 L_s^2\sigma^2}{128\tau}(1+\epsilon_\eta)\right). \tag{176}$$

Using $\tilde{\eta}^{(t)} \leq \frac{1}{4\tilde{L}_s}$ and $\frac{\eta}{\epsilon + \frac{N w_{\max}}{4}} \leq \tilde{\eta}^{(t)} \leq \frac{\eta}{\epsilon}$ we can further simplify

$$\frac{\eta}{2\left(\epsilon + \frac{N w_{\max}}{4}\right)}\left\|\nabla G(\overline{\mathbf{x}}^{(t,0)})\right\|^2 \leq \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t,0)})\right] - \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t+1,0)})\right] + \frac{2\eta N \tilde{L}_g^2 \Xi_t^2}{\epsilon} + \eta^2\left(\frac{\tilde{L}_s w_{\max}^2 \sigma^2 (1+\epsilon_\eta)}{8\tau N \epsilon^2}\right)$$
$$+ \eta\left(\frac{2L_c^2\epsilon_b}{\epsilon} + \frac{L_c^2 w_{\max}^2}{8\epsilon}\epsilon_\eta\right) + \eta^3\left(\frac{w_{\max}^4 L_s^2 L_c^2}{128\epsilon^3}(1+\epsilon_\eta) + \frac{3w_{\max}^4 L_s^2\sigma^2}{128\tau\epsilon^3}(1+\epsilon_\eta)\right). \tag{177}$$

Averaging over T and using lemma 16 alongwith the fact $\tilde{\eta}^{(t)} \leq \frac{\eta}{\epsilon}$ we have

$$\frac{\eta}{2\left(\epsilon + \frac{N w_{\max}}{4}\right)}\frac{1}{T}\sum_{t=1}^{T}\left\|\nabla G(\overline{\mathbf{x}}^{(t,0)})\right\|^2 \leq \frac{G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}}{T} + \eta^2\left(\frac{\tilde{L}_s w_{\max}^2 \sigma^2 (1+\epsilon_\eta)}{8\tau N\epsilon^2}\right) + \eta\left(\frac{4N\tilde{L}_g\Xi_0^2}{\epsilon p T} + \frac{2L_c^2\epsilon_b}{\epsilon} + \right.$$
$$\left. \frac{L_c^2 w_{\max}^2}{8\epsilon}\epsilon_\eta\right) + \eta^3\left(\frac{w_{\max}^4 L_s^2 L_c^2(1+\epsilon_\eta)}{128\epsilon^3} + \frac{3w_{\max}^4 L_s^2\sigma^2(1+\epsilon_\eta)}{128\tau\epsilon^3} + \frac{3N\tilde{L}_g^2 w_{\max}^2 L_c^2(1+\epsilon_\eta)}{2p^2T\epsilon^3} + \frac{3N\tilde{L}_g^2 w_{\max}^2\sigma^2(1+\epsilon_\eta)}{p^2\tau T\epsilon^3}\right). \tag{178}$$

Choosing $\epsilon = \frac{\gamma N B_1}{L_s}$ where $B_1 = \frac{\lambda_{\max} C_{\max} K_{\max} L_c^2}{16} + \frac{\lambda_{\max} K_{\max} L_c^2}{6\sqrt{3}}$(see lemma 8) to make the step size inversely proportional to the global smoothness parameter

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G(\overline{\mathbf{x}}^{(t,0)})\right\|^2\right] \leq \frac{A_0}{\eta T} + A_1 + A_2\eta + A_3\eta^2 \tag{179}$$

where

$$A_0 = 2N\left(\frac{\gamma B_1}{L_s} + \frac{w_{\max}}{4}\right)\left[G(\overline{\mathbf{x}}^{(0,0)}) - G_{\inf}\right], \tag{180}$$

$$A_1 = 2\left(\gamma B_1 + \frac{L_s w_{\max}}{4}\right)\left(\frac{4N\tilde{L}_g\Xi_0^2}{\gamma B_1 p T} + \frac{2L_c^2\epsilon_b}{\gamma B_1} + \frac{L_c^2 w_{\max}^2}{8\gamma B_1}\epsilon_\eta\right), \tag{181}$$

$$A_2 = \frac{\tilde{L}_s\, w_{\max}^2\sigma^2(1+\epsilon_\eta)L_s}{4\tau N^2\gamma^2 B_1^2}\left(\gamma B_1 + \frac{L_s w_{\max}}{4}\right), \tag{182}$$

$$A_3 = \left(\gamma B_1 + \frac{L_s w_{\max}}{4}\right)\left[\frac{w_{\max}^4 L_s^4 L_c^2(1+\epsilon_\eta)}{64N^2\gamma^3 B_1^3} + \frac{3w_{\max}^4 L_s^4\sigma^2(1+\epsilon_\eta)}{64\tau N^2\gamma^3 B_1^3}\right.$$
$$\left. + \frac{3\tilde{L}_g^2 w_{\max}^2 L_c^2(1+\epsilon_\eta)L_s^2}{p^2 T N\gamma^3 B_1^3} + \frac{6\tilde{L}_g^2 w_{\max}^2\sigma^2(1+\epsilon_\eta)L_s^2}{p^2\tau T N\gamma^3 B_1^3}\right]. \tag{183}$$

For small values of gamma we can use the fact $\tilde{\eta}^{(t)} \leq \frac{\eta}{N(1-\gamma)}$ and using the same steps as above starting from equation (45) we obtain the bound

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right] \leq \frac{A_0}{\eta T} + A_1 + A_2\eta + A_3\eta^2 \tag{184}$$

for

$$A_0 = 2N\left(\frac{\gamma B_1}{L_s} + \frac{w_{\max}}{4}\right)\left[G(\overline{\mathbf{x}}^{(0,0)}) - G_{\inf}\right], \tag{185}$$

$$A_1 = 2\left(\frac{\gamma B_1}{L_s} + \frac{w_{\max}}{4}\right)\left(\frac{4N\tilde{L}_g\Xi_0^2}{(1-\gamma)pT} + \frac{2L_c^2\epsilon_b}{(1-\gamma)} + \frac{L_c^2 w_{\max}^2}{8(1-\gamma)}\epsilon_\eta\right), \tag{186}$$

$$A_2 = \frac{\tilde{L}_s\, w_{\max}^2\sigma^2(1+\epsilon_\eta)}{4\tau N^2(1-\gamma)^2}\left(\frac{\gamma B_1}{L_s} + \frac{w_{\max}}{4}\right) \tag{187}$$

$$A_3 = \left(\frac{\gamma B_1}{L_s} + \frac{w_{\max}}{4}\right)\left[\frac{w_{\max}^4 L_s^2 L_c^2(1+\epsilon_\eta)}{64N^2(1-\gamma)^3} + \frac{3w_{\max}^4 L_s^2\sigma^2(1+\epsilon_\eta)}{64\tau N^2(1-\gamma)^3}\right.$$
$$\left. + \frac{3\tilde{L}_g^2 w_{\max}^2 L_c^2(1+\epsilon_\eta)}{p^2TN(1-\gamma)^3} + \frac{6\tilde{L}_g^2 w_{\max}^2\sigma^2(1+\epsilon_\eta)}{p^2\tau TN(1-\gamma)^3}\right]. \tag{188}$$

The final bound is obtained by taking the minimum of the above two expressions and using lemma 1. Specifically choosing $A_4 = \max\left\{\frac{4\tilde{L}_s}{\epsilon}, \frac{3}{\sqrt{2}\epsilon}\right\}$ we have for $\epsilon = \frac{\gamma N B_1}{L_s}$ chosen as per lemma (8).

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right] \leq A_1 + 2\left(\frac{A_2A_0}{T}\right)^{\frac{1}{2}} + 2A_3^{\frac{1}{3}}\left(\frac{A_0}{T}\right)^{\frac{2}{3}} + \frac{A_4A_0}{T} \tag{189}$$

$$= \mathcal{O}\left(\epsilon_\eta\right) + \mathcal{O}\left(\epsilon_b\right) + \mathcal{O}\left(\sqrt{\frac{\left(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\right)(1+\epsilon_\eta)}{N\tau T}}\right)$$
$$+ \mathcal{O}\left(\frac{N\Xi_0^2}{pT}\right) + \mathcal{O}\left(\frac{(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf})}{T}\right)$$
$$+ \mathcal{O}\left(\frac{\left(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\right)^{2/3}}{T^{2/3}}(1+\epsilon_\eta)^{1/3}\right) + \mathcal{O}\left(\frac{\left(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\right)^{2/3}N^{1/3}}{p^{2/3}T^{2/3}}(1+\epsilon_\eta)^{1/3}\right)$$
$$+ \mathcal{O}\left(\frac{\left(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\right)^{2/3}}{\tau^{1/3}T^{2/3}}(1+\epsilon_\eta)^{1/3}\right) + \mathcal{O}\left(\frac{\left(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\right)^{2/3}N^{1/3}}{p^{2/3}\tau^{1/3}T^{2/3}}(1+\epsilon_\eta)^{1/3}\right) \tag{190}$$

D.1 Auxiliary Lemmas

LEMMA 8 (SMOOTHNESS OF THE GLOBAL OBJECTIVE). *Under Assumption 2 the smoothness parameter of the global objective $G(\mathbf{x})$ for any $\mathbf{x}$ is given by $\gamma B_1 + \frac{L_s}{N}\sum_{i=1}^N\left((1-\gamma) + \gamma\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x}))\right)$ where $B_1 = \frac{\lambda_{\max}C_{\max}K_{\max}L_c^2}{16} + \frac{\lambda_{\max}K_{\max}L_c^2}{6\sqrt{3}}$.*

PROOF. From equation (133) have

$$\nabla G(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N\left((1-\gamma) + \gamma\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x}))\right)\nabla F_i(\mathbf{x}). \tag{191}$$

Let $\nabla G_1(\mathbf{x}) := \frac{(1-\gamma)}{N}\sum_{i=1}^N F_i(\mathbf{x})$ and $\nabla G_2(\mathbf{x}) := \frac{\gamma}{N}\sum_{i=1}^N\left\{\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x})\nabla F_i(\mathbf{x})\right\}$. We have

$$\left\|\nabla^2 G_2(\mathbf{x})\right\|_{op} = \gamma\left\|\frac{1}{N}\sum_{i=1}^N\sum_{k\in\mathcal{K}_i}\lambda_k\left\{\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x}))\nabla^2F_i(\mathbf{x})\right.\right.$$
$$+ \sum_{\substack{l\in C_k\\ i\neq l}}\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_l(\mathbf{x}))(1-\sigma_i(\mathbf{x}))\nabla F_l(\mathbf{x})\nabla F_i(\mathbf{x})^T$$

$$\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x}))(1-2\sigma_i(\mathbf{x}))\nabla F_i(\mathbf{x})\nabla F_i(\mathbf{x})^T\Bigg\|_{op}. \tag{192}$$

Using triangle inequality for operator norms and $\sigma(x)(1-\sigma(x)) \le 1/4$ and $\sigma(x)(1-\sigma(x))(1-2\sigma(x)) \le 1/6\sqrt{3}$ we have

$$\left\|\nabla^2 G_2(\mathbf{x})\right\|_{op} \le \frac{\gamma}{N}\sum_{i=1}^{N}\sum_{k\in\mathcal{K}_i}\lambda_k\left\{L_s\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x}))+\sum_{\substack{l\in C_k\\ i\neq l}}\frac{L_c^2}{16}+\frac{L_c^2}{6\sqrt{3}}\right\} \tag{193}$$

$$\le \frac{\gamma\lambda_{\max}C_{\max}K_{\max}L_c^2}{16}+\frac{\gamma\lambda_{\max}K_{\max}L_c^2}{6\sqrt{3}}+\frac{\gamma L_s}{N}\sum_{i=1}^{N}\left\{\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x}))\right\}. \tag{194}$$

Also $\|\nabla^2 G_1(\mathbf{x})\|_{op} \le (1-\gamma)L_s$. Since $\|\nabla^2 G(\mathbf{x})\|_{op} \le \|\nabla^2 G_1(\mathbf{x})\|_{op} + \|\nabla^2 G_2(\mathbf{x})\|_{op}$

$$\|\nabla^2 G(\mathbf{x})\|_{op} \le \gamma B_1 + \frac{L_s}{N}\sum_{i=1}^{N}\left((1-\gamma)+\gamma\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x}))\right) \tag{195}$$

where $B_1 = \frac{\lambda_{\max}C_{\max}K_{\max}L_c^2}{16} + \frac{\lambda_{\max}K_{\max}L_c^2}{6\sqrt{3}}$. □

LEMMA 9 (SMOOTHNESS OF Ψ). *For all $i \in [N]$ we have $\|[\nabla\Psi(\mathbf{X})]_i - [\Psi(\mathbf{Y})]_i\|_2^2 \le \frac{\tilde{L}_g}{N}\|\mathbf{X}-\mathbf{Y}\|_F^2$ where $\tilde{L}_g = \frac{L_s w_{\max}}{4} + \frac{L_c^2 w_{\max}}{4}$.*

PROOF. From equation (136) we have

$$[\nabla\Psi(\mathbf{X})]_i = \nabla_{\mathbf{x}_i}\Psi(\mathbf{X}) = \frac{1}{N}\left((1-\gamma)+\gamma\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x}_j)\right)(1-\sigma_i(\mathbf{x}_i))\right)\nabla F_i(\mathbf{x}_i). \tag{196}$$

We also have

$$\nabla^2_{\mathbf{x}_i}\Psi(\mathbf{X}) = \frac{1}{N}\left\{\left((1-\gamma)+\gamma\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x}_j)\right)(1-\sigma_i(\mathbf{x}_i))\right)\nabla^2 F_i(\mathbf{x}_i)\right. \tag{197}$$

$$\left.+\left(\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x}_j)\right)(1-\sigma_i(\mathbf{x}_i))(1-2\sigma_i(\mathbf{x}_i))\right)\nabla F_i(\mathbf{x}_i)\nabla F_i(\mathbf{x}_i)^\top\right\} \tag{198}$$

and for $i \neq l$ we have

$$\nabla^2_{\mathbf{x}_i,\mathbf{x}_l}\Psi(\mathbf{X}) = \frac{1}{N}\sum_{k\in\mathcal{K}_i\cap\mathcal{K}_l}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x}_j)\right)(1-\sigma_i(\mathbf{x}_i))(1-\sigma_l(\mathbf{x}_l))\nabla F_i(\mathbf{x}_i)\nabla F_l(\mathbf{x}_l)^\top. \tag{199}$$

Using triangle inequality for operator norms and $\sigma(x)(1-\sigma(x)) \le 1/4$ and $\sigma(x)(1-\sigma(x))(1-2\sigma(x)) \le 1/6\sqrt{3}$ we have

$$\|\nabla^2_{\mathbf{x}_i}\Psi(\mathbf{X})\|_{op} \le \frac{L_s(4(1-\gamma)+\gamma\sum_{k\in\mathcal{K}_i}\lambda_k)}{4N}+\frac{L_c^2(6\sqrt{3}(1-\gamma)+\gamma\sum_{k\in\mathcal{K}_i}\lambda_k)}{6\sqrt{3}N} \le \frac{L_s w_{\max}}{4N}+\frac{L_c^2 w_{\max}}{4N} \tag{200}$$

$$\|\nabla^2_{\mathbf{x}_i,\mathbf{x}_l}\Psi(\mathbf{X})\|_{op} \le \frac{L_c^2(\sum_{k\in\mathcal{K}_i\cap\mathcal{K}_j}\lambda_i)}{6\sqrt{3}N}. \tag{201}$$

Now by the mean value theorem for any $\mathbf{X}$ and $\mathbf{Y}$, we have $[\nabla\Psi(\mathbf{X})]_i - [\nabla\Psi(\mathbf{Y})]_i = \sum_{j=1}^N \nabla^2_{x_i,x_j}\Psi(\mathbf{Z})(x_j-y_j)$ for some intermediate point $\mathbf{Z} = t\mathbf{X}+(1-t)\mathbf{Y}$ where $t\in[0,1]$. A uniform upper bound on these operator norms for all i can be given by $\tilde{L}_g/N$ where $\tilde{L}_g = \frac{L_s w_{\max}}{4}+\frac{L_c^2 w_{\max}}{4}$ and thus $\|\nabla^2_{x_i,x_j}\Psi(\mathbf{Z})\|_{op} \le \tilde{L}_g/N$, we obtain $\|[\nabla\Psi(\mathbf{X})]_i - [\nabla\Psi(\mathbf{Y})]_i\| \le \tilde{L}_g/N\sum_{j=1}^N\|x_j-y_j\|$. Using Jensen's inequality we obtain

$$\|[\nabla\Psi(\mathbf{X})]_i - [\Psi(\mathbf{Y})]_i\|_2^2 \le \frac{\tilde{L}_g^2}{N}\sum_{j=1}^{N}\|x_j-y_j\|^2. \tag{202}$$

□

LEMMA 10 (PRE-FACTOR UPPER BOUND). *For $w_{\max} = 4(1-\gamma)+\gamma\max_{i\in[N]}\left(\sum_{k\in\mathcal{K}_i}\lambda_k\right)$, we have $\hat{b}_i^{(t)} \le \frac{w_{\max}}{4}$ and $b_i^{(t)} \le \frac{w_{\max}}{4}$ $\forall i\in[N]$ and t.*

PROOF. We have

$$\hat{b}_i^{(t)} = (1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k M_{ki}(\mathbf{X}^{(t,0)})(1-\sigma_i(\mathbf{x}_i^{(t,0)})).$$

By definition, it follows that

$$\left(M_{ki}^{(0)}(\mathbf{X}^{(t,0)})\right)^{1/N} = \sigma_i(\mathbf{x}_i^{(t,0)})$$

and

$$M_{ki}(\mathbf{X}^{(t,0)}) = \min\left\{\sigma_i(\mathbf{x}_i^{(t,0)}), M_{ki}^{(\tau_b)}(\mathbf{X}^{(t,0)})\right\}.$$

Therefore, we obtain the bound

$$\hat{b}_i^{(t)} \le (1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k \sigma_i(\mathbf{x}_i^{(t,0)})(1-\sigma_i(\mathbf{x}_i^{(t,0)})) \le (1-\gamma) + \frac{\gamma \sum_{k\in\mathcal{K}_i}\lambda_k}{4} \le \frac{4(1-\gamma) + \gamma \max_{i\in[N]}\left(\sum_{k\in\mathcal{K}_i}\lambda_k\right)}{4}.$$

We also have $i \in \mathcal{K}_i$ by definition thus $M_k(\mathbf{X}^{(t,0)}) \le \sigma_i(\mathbf{x}_\mathbf{i}^{(\mathbf{t},\mathbf{0})})$ and

$$b_i^{(t)} \le (1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k \sigma_i(\mathbf{x}_i^{(t,0)})(1-\sigma_i(\mathbf{x}_i^{(t,0)})) \le (1-\gamma) + \frac{\gamma \sum_{k\in\mathcal{K}_i}\lambda_k}{4} \le \frac{4(1-\gamma) + \gamma \max_{i\in[N]}\left(\sum_{k\in\mathcal{K}_i}\lambda_k\right)}{4}.$$

□

LEMMA 11 (LEARNING RATE ESTIMATION ERROR - GROUPWISE PENALTY). *For τ_η rounds of gossip averaging used for estimating $s_i^{(t)}$ we have*

$$\frac{1}{N}\sum_{i=1}^{N}\left\|\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right\|^2 (\hat{b}_i^{(t)})^2 \le \min\left\{\frac{(\eta)^2(1-p)^{\tau_\eta}N^2}{4}, \frac{w_{\max}^2(\tilde{\eta}^{(t)})^2(1-p)^{\tau_\eta}N^2}{64}\right\} \tag{203}$$

$$= \min\left\{(\eta)^2\epsilon_\eta, \frac{(\tilde{\eta}^{(t)})^2 w_{\max}^2 \epsilon_\eta}{16}\right\}. \tag{204}$$

where $\epsilon_\eta = \frac{(1-p)^{\tau_\eta}N^2}{4}$. Note that ϵ_η can be made arbitrarily small as required by increasing τ_η.

PROOF. We have $\tilde{\eta}_i^{(t)} = \frac{\eta}{s_i^{(t)}+\epsilon}$ and $\tilde{\eta}^{(t)} = \frac{\eta}{\bar{s}^{(t)}+\epsilon}$ where $s_i^{(t)} = \max\{s_i^{(t,\tau_\eta)}, s_i^{(t,0)}/N\}$, $s_i^{(t,0)} = N\hat{b}_i^{(t)}$, and $s_i^{(t,r+1)} = \sum_{j\in\mathcal{N}_i} w_{ij}\, s_j^{(t,r)}$, or equivalently $\mathbf{s}^{(t,r+1)} = \mathbf{W}\mathbf{s}^{(t,r)}$, with $\mathbf{s}^{(t,r)} = [s_1^{(t,r)} \cdots s_N^{(t,r)}]$. The proof follows same as lemma 3 after replacing $b_i^{(t)}$ by $\hat{b}_i^{(t)}$ and using $\hat{b}_i^{(t)} \le \frac{w_{\max}}{4}$ from lemma 10. □

LEMMA 12 (FRACTIONAL LEARNING RATE ESTIMATION ERROR - GROUPWISE PENALTY). *For τ_η rounds of gossip averaging used for estimating $s_i^{(t)}$ we have*

$$\frac{1}{N}\sum_{i=1}^{N}\frac{\left\|\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right\|^2}{\|\tilde{\eta}^{(t)}\|^2}(\hat{b}_i^{(t)})^2 \le \frac{(1-p)^{\tau_\eta}N^2(\max_i(\hat{b}_i^{(t)}))^2}{4} = \max_i(\hat{b}_i^{(t)})^2\epsilon_\eta \le \frac{w_{\max}^2\epsilon_\eta}{16}. \tag{205}$$

PROOF. Proceeding similarly as lemma 3 replacing $b_i^{(t)}$ by $\hat{b}_i^{(t)}$ and using lemma 10 for bounding $\hat{b}_i^{(t)}$ completes the proof. □

LEMMA 13 (PRODUCT ESTIMATION ERROR). *If $M_{ki}(\mathbf{X})$ is the estimate of $M_k(\mathbf{X})$ at node i computed as per equation* (137)*, then for any vector $\mathbf{X}$ we have*

$$\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}\lambda_k^2\,\|M_k(\mathbf{X}) - M_{ki}(\mathbf{X})\|^2 \le (1-p)^{\tau_b}(N-1)\sum_{k=1}^{K}\lambda_k^2 = \epsilon_M. \tag{206}$$

Note that this term can be made arbitrarily small by increasing τ_b.

PROOF. We have $M_k(\mathbf{X}) := \prod_{j\in C_k}\sigma_j(\mathbf{x}_j)$ and $M_{ki}(\mathbf{X}) = \min\left\{M_{ki}^{(\tau_b)}(\mathbf{X}), (M_{ki}^{(0)}(\mathbf{X}))^{1/N}\right\}$. Note that $M_k(\mathbf{X}) \le (M_{ki}^{(0)}(\mathbf{X}))^{1/N}$ $\forall i$ and therefore $\|M_k(\mathbf{X}) - M_{ki}(\mathbf{X})\|^2 \le \left\|M_k(\mathbf{X}) - M_{ki}^{(\tau_b)}(\mathbf{X})\right\|^2$. Now we have

$$\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}\lambda_k^2\,\|M_k(\mathbf{X}) - M_{ki}(\mathbf{X})\|^2 \le \frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}\lambda_k^2\left\|M_k(\mathbf{X}) - M_{ki}^{(\tau_b)}\right\|^2 \tag{207}$$

$$= \frac{1}{N}\sum_{k=1}^{K}\sum_{i=1}^{N}\lambda_k^2\left\|M_k(\mathbf{X}) - M_k(\mathbf{X})^{\frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))}{\ln(M_k(\mathbf{X}))}}\right\|^2 \tag{208}$$

$$= \frac{1}{N}\sum_{k=1}^{K}\sum_{i=1}^{N}\lambda_k^2 \left\| M_k(\mathbf{X}) - M_k(\mathbf{X})^{1+\frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))-\ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))}} \right\|^2. \tag{209}$$

For a positive ϵ the inequality, $a - a^{1+\epsilon} \leq \frac{\epsilon}{(1+\epsilon)^{1+\frac{1}{\epsilon}}} \leq \frac{\epsilon}{1+\epsilon} \leq \epsilon$ holds for $a \in [0,1]$. Since both $M_k(\mathbf{X})$ and $M_{ki}^{(\tau_b)}(\mathbf{X}) \in [0,1]$, if $\frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))-\ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))} > 0$ in equation(209) for $\epsilon = \frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))-\ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))}$

$$\frac{1}{N}\sum_{k=1}^{K}\sum_{i=1}^{N}\lambda_k^2 \left\| M_k(\mathbf{X}) - M_k(\mathbf{X})^{1+\frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))-\ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))}} \right\|^2 \leq \frac{1}{N}\sum_{k=1}^{K}\sum_{i=1}^{N}\lambda_k^2 \left\| \frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X})) - \ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))} \right\|^2. \tag{210}$$

For the case where $\frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))-\ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))} < 0$ we can do the same procedure using $M_{ki}^{(\tau_b)}(\mathbf{X})$ as the base, here the ϵ would be $\frac{\ln(M_k(\mathbf{X}))-\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))}{\ln(M_{ki}(\mathbf{X}))}$ and we upper bound using $\frac{\epsilon}{1+\epsilon}$ to obtain $\left\| M_k(\mathbf{X}) - M_{ki}^{(\tau_b)}(\mathbf{X}) \right\|^2 \leq \left\| \frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))-\ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))} \right\|^2$. Since the numerator in this expression corresponds exactly to the squared norm of a scaled version of the object being computed over the network via gossip. from assumption 4, after τ_b rounds of gossip with the mixing matrix $\mathbf{W}$ we obtain

$$\frac{1}{N}\sum_{k=1}^{K}\sum_{i=1}^{N}\lambda_k^2 \left\| \frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X})) - \ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))} \right\|^2 = \frac{1}{N}\sum_{k=1}^{K}\lambda_k^2 \left(\frac{\sum_{i=1}^{N}\left\| \ln(M_{ki}^{(\tau_b)}(\mathbf{X})) - \ln(M_k(\mathbf{X})) \right\|^2}{\|\ln(M_k(\mathbf{X}))\|^2} \right) \tag{211}$$

$$\leq \frac{1}{N}\sum_{k=1}^{K}\lambda_k^2 \left(\frac{(1-p)^{\tau_b}\sum_{i=1}^{N}\left\| \ln(M_{ki}^{(0)}(\mathbf{X})) - \ln(M_k(\mathbf{X})) \right\|^2}{\|\ln(M_k(\mathbf{X}))\|^2} \right) \tag{212}$$

$$= \frac{1}{N}\sum_{k=1}^{K}\lambda_k^2 \left((1-p)^{\tau_b}\sum_{i=1}^{N}\left\| \frac{\ln(M_{ki}^{(0)}(\mathbf{X})) - \ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))} \right\|^2 \right) \tag{213}$$

$$\leq \frac{1}{N}\sum_{k=1}^{K}\lambda_k^2 (1-p)^{\tau_b}(N^2 - N) \tag{214}$$

$$= (1-p)^{\tau_b}(N-1)\sum_{k=1}^{K}\lambda_k^2 = \epsilon_M. \tag{215}$$

Additionally, Equation (213) exhibits a structure similar to the coefficient of variation. Equation (215) follows from the fact that for non-negative weights w_i and the mean weight $w = \frac{1}{N}\sum_{i=1}^{N} w_i$, the following inequality holds:

$$\sum_{i=1}^{N}\frac{\|w_i - w\|^2}{\|w\|^2} = \frac{\sum_{i=1}^{N}\|w_i\|^2}{\|\frac{1}{N}\sum_{i=1}^{N} w_i\|^2} - N \leq N^2 - N.$$

□

LEMMA 14 (PRE-FACTOR ESTIMATION ERROR). *The pre-factor estimation error after τ_b rounds of gossip averaging used for estimating the product terms $M_k(\mathbf{X}^{(t,0)})$ can be upper bounded as*

$$\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\left[\left\| b_i^{(t)} - \hat{b}_i^{(t)} \right\|^2 \right] \leq K_{\max}\epsilon_M = \epsilon_b. \tag{216}$$

PROOF.

$$\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\left[\left\| b_i^{(t)} - \hat{b}_i^{(t)} \right\|^2 \right] = \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\left[\left\| \sum_{k\in\mathcal{K}_i}\lambda_k \left(M_k(\mathbf{X}^{(t,0)}) - M_{ki}(\mathbf{X}^{(t,0)}) \right) \right\|^2 (1-\sigma_i(\mathbf{x}_i^{(t,0)}))^2 \right] \tag{217}$$

$$\leq \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\left[\sum_{k\in\mathcal{K}_i}|\mathcal{K}_i|\lambda_k^2 \left\| \left(M_k(\mathbf{X}^{(t,0)}) - M_{ki}(\mathbf{X}^{(t,0)}) \right) \right\|^2 \right] \tag{218}$$

$$\leq \frac{K_{\max}}{N}\mathbb{E}\left[\sum_{k=1}^{K}\sum_{i=1}^{N}\lambda_k^2 \left\| \left(M_k(\mathbf{X}^{(t,0)}) - M_{ki}(\mathbf{X}^{(t,0)}) \right) \right\|^2 \right] \tag{219}$$

$$\leq K_{\max}\epsilon_M = \epsilon_b \tag{220}$$

□

LEMMA 15 (BOUNDING DRIFT - GROUPWISE PENALTY). *For client i, the accumulated local drift error over τ local steps in Algorithm 3, can be bounded as:*

$$\sum_{r=1}^{\tau}\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] \leq \tau L_c^2\left((\tilde{\eta}^{(t)})^2(\hat{b}_i^{(t)})^2+\mathbb{E}\left[\left\|(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\right\|^2\right]\right) \frac{3(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{16}+3\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2\right]. \tag{221}$$

PROOF. Using the expression for the accumulated local update after r local steps:

$$\tau^2\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] = \mathbb{E}\left[\left\|\tilde{\eta}_i^{(t)}\,\hat{b}_i^{(t)}\sum_{l=0}^{r-1}\nabla F_i\left(\mathbf{x}_i^{(t,l)},\xi_i^{(t,l)}\right)\right\|^2\right] \tag{222}$$

$$\leq 3\mathbb{E}\left[\left\|\tilde{\eta}^{(t)}\,\hat{b}_i^{(t)}\sum_{l=0}^{r-1}\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right]+3\mathbb{E}\left[\left\|(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\sum_{l=0}^{r-1}\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right] + \underbrace{3\mathbb{E}\left[\left\|(\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\sum_{l=0}^{r-1}\left(\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)-\nabla F_i\left(\mathbf{x}_i^{(t,l)},\xi_i^{(t,l)}\right)\right)\right\|^2\right]}_{T_{17}}. \tag{223}$$

Consider T_{17}

$$T_{17} = 3\mathbb{E}\left[\left\|(\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\sum_{l=0}^{r-1}\left(\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)-\nabla F_i\left(\mathbf{x}_i^{(t,l)},\xi_i^{(t,l)}\right)\right)\right\|^2\right] \tag{224}$$

$$\leq 3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2\,(\hat{b}_i^{(t)})^2\sum_{l=0}^{r-1}\left\|\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)-\nabla F_i\left(\mathbf{x}_i^{(t,l)},\xi_i^{(t,l)}\right)\right\|^2\right] \tag{225}$$

$$\leq 3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2\,(\hat{b}_i^{(t)})^2 r\sigma^2\right] \tag{226}$$

$$\leq 6\mathbb{E}\left[(\tilde{\eta}^{(t)})^2(\hat{b}_i^{(t)})^2 r\sigma^2\right]+6\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2 r\sigma^2\right] \tag{227}$$

$$\leq \frac{3(\tilde{\eta}^{(t)})^2 r w_{\max}^2\sigma^2}{8}+6r\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2\right]. \tag{228}$$

Here equation (225) follows from proposition 4 and equation 228 follows from lemma 10. Plugging this back into equation 223 we have

$$\tau^2\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] \leq 3\mathbb{E}\left[\left\|\tilde{\eta}^{(t)}\,\hat{b}_i^{(t)}\sum_{l=0}^{r-1}\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right] + 3\mathbb{E}\left[\left\|(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\sum_{l=0}^{r-1}\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right]+\frac{3(\tilde{\eta}^{(t)})^2 r w_{\max}^2\sigma^2}{8}+6r\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2\right] \tag{229}$$

$$\leq 3r\sum_{l=1}^{r-1}\left(\mathbb{E}\left[\left\|\tilde{\eta}^{(t)}\,\hat{b}_i^{(t)}\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right]+\mathbb{E}\left[\left\|(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right]\right) + \frac{3(\tilde{\eta}^{(t)})^2 r w_{\max}^2\sigma^2}{8}+6r\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2\right] \tag{230}$$

$$\leq 3r^2L_c^2\left((\tilde{\eta}^{(t)})^2(\hat{b}_i^{(t)})^2+\mathbb{E}\left[\left\|(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\right\|^2\right]\right)+\frac{3(\tilde{\eta}^{(t)})^2 r w_{\max}^2\sigma^2}{8}+6r\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2\right]. \tag{231}$$

Summing over r and dividing by τ^2 we have

$$\sum_{r=1}^{\tau}\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] \leq \tau L_c^2\left((\tilde{\eta}^{(t)})^2(\hat{b}_i^{(t)})^2+\mathbb{E}\left[\left\|(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\right\|^2\right]\right)$$

$$\frac{3(\tilde{\eta}^{(t)})^2 w_{\max}^2 \sigma^2}{16} + 3\mathbb{E}\left[(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})^2 (\hat{b}_i^{(t)})^2 \sigma^2\right]. \tag{232}$$

□

LEMMA 16 (CONSENSUS CONTROL- GROUPWISE PENALTY). *The term* $\frac{1}{T}\sum_{t=0}^{T-1}\Xi_t^2$ *can be bounded as*

$$\frac{1}{T}\sum_{t=0}^{T-1}\Xi_t^2 \le \frac{2\Xi_0^2}{pT} + \left(\frac{3w_{\max}^2 L_c^2}{4p^2 T} + \frac{3w_{\max}^2\sigma^2}{2p^2\tau T}\right)(1+\epsilon_\eta)\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2. \tag{233}$$

(234)

PROOF. Proceeding similar to the proof of lemma 6 replacing $q_i(\mathbf{x}_i^{(t,0)})a_i$ by $\hat{b}_i^{(t)}$ we have

$$N\Xi_{t+1}^2 \le \left(1-\frac{p}{2}\right)N\Xi_t^2 + \underbrace{\frac{2}{p\tau^2}\sum_{i=1}^{N}\mathbb{E}\left\|\tilde{\eta}_i^{(t)}\hat{b}_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)},\xi_i^{(t,r)})\right\|^2}_{T_{18}}. \tag{235}$$

Following the same steps as equations (222)-(231), replacing r with τ we bound T_{18} as

$$\begin{aligned} T_{18} \le \frac{2}{p\tau^2}\sum_{i=1}^{N}\Bigg(&3\tau^2 L_c^2\left((\tilde{\eta}^{(t)})^2(\hat{b}_i^{(t)})^2 + \mathbb{E}\left[\left\|(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\right\|^2\right]\right) \\ &+ \frac{3(\tilde{\eta}^{(t)})^2\tau w_{\max}^2\sigma^2}{8} + 6\tau\mathbb{E}\left[(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2\right]\Bigg). \end{aligned} \tag{236}$$

Now using lemma 10 and 11 we can bound (236) as

$$T_{18} \le \frac{3(\tilde{\eta}^{(t)})^2 N w_{\max}^2 L_c^2}{8p}(1+\epsilon_\eta) + \frac{3(\tilde{\eta}^{(t)})^2 N w_{\max}^2\sigma^2}{4p\tau}(1+\epsilon_\eta) = NA_t. \tag{237}$$

Thus we have

$$\Xi_{t+1}^2 \le \left(1-\frac{p}{2}\right)\Xi_t^2 + A_t. \tag{238}$$

Rearranging and summing over T we get

$$\Xi_T^2 - \Xi_0^2 = \sum_{t=0}^{T-1}\left(\Xi_{t+1}^2 - \Xi_t^2\right) \le -\frac{p}{2}\sum_{t=0}^{T-1}\Xi_t^2 + \sum_{t=0}^{T-1}A_t. \tag{239}$$

Since $\Xi_T^2 \ge 0$

$$\frac{1}{T}\sum_{t=0}^{T-1}\Xi_t^2 \le \frac{2}{pT}\left(\Xi_0^2 + \sum_{t=0}^{T-1}A_t\right). \tag{240}$$

Substituting A_t completes the proof. □

E Additional experimental details

Value of γ for replication. Note that a scaled version of our objective function can be rewritten as $F(\mathbf{x}) + \beta\tilde{F}(\mathbf{x})$ where $\tilde{F}(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^{N} a_i\sigma_i(\mathbf{x})$ and $\beta = \frac{\gamma}{1-\gamma}$. As the heterogeneity in data increases we typically need larger values of β to retain more clients. A large value for β can be used here because the sigmoid function decays exponentially once a client's loss falls below the threshold and the objective is dominated by F. For all our experiments $\beta = 20$.

Hyperparameters and local steps for replication. We use a batch size of 32 for all experiments. For FashionMNIST and EMNIST, we train for $T = 300$ communication rounds with $\tau = 5$ local SGD steps per round. For CIFAR-10 and CelebA, due to complexity and training time, we train for $T = 120$ rounds with the same $\tau = 5$ local steps. In NodeDrop-IDSGD, we set the base learning rate to 1 (the resulting effective learning rates are much smaller due to the adaptive scaling) and rely on adaptive per-client scaling factors to determine the effective step sizes, whereas the decentralized SGD baselines use fixed learning rates of 0.005 (FashionMNIST, EMNIST) and 0.0005 (CIFAR-10, CelebA). We set the number of gossip averaging steps used to estimate the scaling factors (τ_{st}) and pre-factors ($\tau_{\log}$) to 5 in all settings, and report results averaged over 10 random seeds.

Architectures for replication. For FashionMNIST we use a deep MLP with two hidden layers containing 64 and 30 units, respectively, with a dropout layer after the first hidden layer; the input is a normalized, flattened 28×28 image and the output has 10 units. For EMNIST we use a LeNet-5 style convolutional network adapted to 28×28 grayscale input with 47 output classes. For CIFAR-10 (and CIFAR-100) we use a three-layer convolutional network with max-pooling, batch normalization, a 256-unit fully connected layer, dropout, and a final classification

layer. For CelebA we use a small convolutional network with three convolution–batch-normalization–max-pooling blocks, followed by global average pooling, dropout, and a 2-unit output layer for the smiling/not-smiling task.

Client thresholds for replication. A practical approach to selecting local loss thresholds for each client involves running a few warm-up steps of local SGD at client k to obtain the model $\widehat{\mathbf{x}}_k$, then using the training loss $F(\widehat{\mathbf{x}}_k)$ of the resulting solo trained model obtained as the threshold. This method provides a data-driven baseline that reflects each client's local data distribution and optimization dynamics. While thresholds can be tailored to client-specific preferences or constraints in real-world settings, for our experiments, we adopt a uniform strategy: each client performs 100 warmup steps to establish its threshold. We use the same fixed learning rate as in the corresponding baseline during the warmup training. This ensures consistency across clients while still capturing their unique local training behaviors.